RESEARCH MONOGRAPH

ON

CLIMATE-ADAPTIVE RIVER MANAGEMENT IN BANGLADESH: A STUDY ON THE CHALLENGES FROM THE LEGAL PERSPECTIVE

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THIS RESEARCH MONOGRAPH IS SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF MASTER OF LAWS (LLM-PROFESSIONALS)

DEPARTMENT OF LAW
FACULTY OF SECURITY & STRATEGIC STUDIES (FSSS)
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TITLE PAGE

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NOVEMBER 2024

DEDICATION

Dedicated to

My Father and Mother who sacrificed the most to our joys

DECLARATION

I, Md. Mehedi Hasan, declare that this research monograph titled "Climate-Adaptive River Management in Bangladesh: A Study on the Challenges from the Legal Perspective", submitted in the fulfilment of the requirements for the degree of Master of Laws (LLM), in the Department of Law, Bangladesh University of Professionals (BUP) is completely outcome of my own work under supervision of Assistant Professor Dr. Md. Abu Hanif. I also declare that this thesis or no part thereof has been or is being submitted elsewhere for the award of my degree.

…….
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Md. Mehedi Hasan

LIST OF ABBREVIATIONS

BCCSAP	Bangladesh Climate Change Strategy and Action Plan
BWDB	Bangladesh Water Development Board
CBD	Convention on Biological Diversity
DoE	Department of Environment
GBM	Ganges-Brahmaputra-Meghna (River Basin)
IWRM	Integrated Water Resource Management
MoWR	Ministry of Water Resources
NAPA	National Adaptation Programme of Action
NRCC	National River Conservation Commission
SDGs	Sustainable Development Goals
UNFCCC	United Nations Framework Convention on Climate Change
WARPO	Water Resources Planning Organization

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CLIMATE-ADAPTIVE RIVER MANAGEMENT IN BANGLADESH: A STUDY ON THE CHALLENGES FROM THE LEGAL PERSPECTIVE

ABSTRACT

This paper examines Bangladesh's legal stance on climate-adaptive river management, assessing the extent to which current laws address climate risks impacting future resilience. Situated in the vulnerable Ganges-Brahmaputra-Meghna basin, Bangladesh faces climate-driven water challenges that affect river ecosystems and dependent communities. Legal frameworks such as the Bangladesh Water Act (2013) and National Water Policy (1999) exist, yet they lack emphasis on climate resilience. This study evaluates these legal tools, identifies regulatory barriers, and proposes reforms for enhancing climate adaptation in river management. Using qualitative analysis of legal documents, including national laws and the UN Watercourses Convention, findings reveal policy gaps and recommend a more cohesive framework to ensure sustainable river management, resource equity, and protection for vulnerable populations in Bangladesh.

Keywords: Climate-Adaptive River Management, Bangladesh Water Law, Climate Resilience, Legal Frameworks, Climate Change Adaptation

CHAPTER I
INTRODUCTION

1.1 Background of the Study

Bangladesh has a unique geography and highly vulnerable to climate change therefore it is one of the most climate-vulnerable countries in the world because its freshwater resources management faces many challenges. Being located at the lower riparian part of the Ganges-Brahmaputra-Meghna (GBM) river basin, Bangladesh is extremely reliant on upstream flow and therefore vulnerable to both natural variability in water availability and anthropogenic interventions in other nations. [1] The vulnerabilities created by rapid urbanization, socio-economic progress, and population growth are worsened significantly by climate change in ways that increase the frequency and severity of extreme weather events including floods, droughts and cyclones which impact river ecosystems as well as water availability with significant implications for millions who rely on these water systems to sustain their livelihoods.[2]

The implication of these impacts has resulted in climate-adaptive river management as a critical necessity. Bangladesh has taken steps to mitigate climate change such as the Bangladesh Water Act（2013）, National Water Policy（1999）and policy putting emphasis on capacity and capability development like Bangladesh Climate Change Strategy and Action Plan (BCCSAP). Still, fragmentation and lack of implementation of the regulatory basis to manage for climate change impact on rivers. Current legislation often emphasizes water conservation and pollution abatement but stops short of addressing climate resilient river management. Without a coordinated, climate responsive legal regime, it is challenged to protect rivers and water resources from the long-term effects of climate change.[3]

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[1] Ahmad Quazi Kholiquzzaman and Ahmed Ainun Nishat, "Regional Cooperation in Flood Management in the Ganges-Brahmaputra-Meghna Region: Bangladesh Perspective," Springer eBooks (2003) <https://doi.org/10.1007/978-94-017-0137-2_9> accessed on 1st October, 2024

[2] Pandey Anil and others, "Upstream-Downstream Linkages in Ganges-Brahmaputra-Meghna Basin: The Hydro-Social Imperatives" (2020) 22 Water Policy 1082 https://doi.org/10.2166/wp.2020.231 accessed on 1st October, 2024

[3] Islam Sheikh Nazrul, "Deltaic Floodplains Development and Wetland Ecosystems Management in the Ganges–Brahmaputra–Meghna Rivers Delta in Bangladesh" (2016) 2 Sustainable Water Resources Management 237 https://doi.org/10.1007/s40899-016-0047-6 accessed on 1st October, 2024

The aim of this study is to examine the performance of Bangladesh legal frameworks in promoting climate adaptive river management, as well as identify a few of the legal barriers that inhibit such adaptation and suggest required reforms. The present study seeks to help fill these gaps and hence contributes to attracting future debate on developing the more climate-resistant river management needed as climate change escalates.

1.2 Identification of Research Problem

Climate change, a grim reality our planet has been debilitated by poses an alarming threat to Bangladesh primarily as it is besieged and edged around entirely with river systems that hold much significance for its socio-economic sustenance coupled with the obvious ecological ramifications. With rising sea levels, more severe and frequent extreme weather events, changes in flow patterns of rivers compound pressures on integrated land clearance/water/flood infrastructure planning to manage climate-adaptative fluvial systems. On the other hand, existing legal frameworks in the country such as Bangladesh Water Act 2013 and National water policy 1999 are inadequate to manage these complexities. Both are inadequate in their inclusion of climate resilience, and both face challenges arising from weak implementation mechanisms as well as blurred institutional tasks. Moreover, it contributes to trans-boundary water management disputes (India in particular) which construe the chronicity of water shortages with Marginalized communities residing at Coastal regions where salinity ingress and climate changing are increasingly affecting their traditional livelihoods. The research focuses on the deficiencies of legal and institutional frameworks governing water in Bangladesh, drawing out reforms which would enable more climate resilient river management through ensuring rights to equitable access for vulnerable populations.

1.3 Research Questions

i. How does Bangladesh's legal framework address climate-adaptive river management?

ii. What are the legal challenges of climate-adaptive river management in Bangladesh?

iii. What legal measures can be taken to strengthen climate-adaptive river management in Bangladesh?

1.4 Objectives of the Research

This research aims to critically examine the legal frameworks and policies governing river management in Bangladesh, with a particular focus on adaptation to climate change. The specific objectives are:

i. To examine Bangladesh's legal framework for addressing climate-adaptive river management.

ii. To identify key legal challenges to climate-adaptive river management in Bangladesh.

iii. To recommend improvements to enhance climate resilience in Bangladesh's river management laws.

1.5 Scope of the Research

The study looks into the river management framework of Bangladesh with particular emphasis on legal and policy dimensions that enable climate decentralisation. The study undertakes a review of national laws, policies and institutions for river governance, with specific focus on their effectiveness in enhancing climate resilience. The study evaluates the status of existing adaptive practices and regulatory impediments in three major river systems, that is Ganges-Brahmaputra-Meghna. It also incorporates certain global frameworks and identifies best practices that can be replicated in the context of Bangladesh. This research attempts to suggest legal and adaptive institutional reforms for effective climate change adaptation in river management; highlighting environmental, economical and social vulnerabilities.

1.6 Limitation of the Research

The primary limitation is that the findings are mainly drawn on secondary sources such as legal and policy, so it perhaps does not reveal in-depth insights about attitudes or experiences. However, because the study is a short-term and limited resource project we are not doing substantial primary data work such as field interviews with stakeholders or even standing up expanded community surveys. Moreover, hydrological and climate modeling for river systems is more complicated than this research seeks to explore. Secondly, the information on climate impacts and adaptive effectiveness are subject to data availability constraints within Bangladesh's river management practices.

1.7 Justification and Significance

The importance of this study lies in the pressing urgency for developing a climate-adaptive river management strategy, especially focusing on Bangladesh, one of the most susceptible countries to global warming. With a focus on legal and policy frameworks, this research assesses key gaps therein suggesting reforms building the adaptive capacity of Bangladesh's river systems to climate stressors such as flooding, droughts and salinity intrusion. Mainstreaming climate adaptation into river management allows the critical ecosystems on which communities depend for their health, livelihoods and socio-economic well-being to be addressed.

This paper aims to fill a research gap in the existing literature by examining legal frameworks that provide support for climate resilience, and offering lessons from international norms best suited to national circumstances. Additionally, it contributes towards environmental justice by campaigning for fair water resource entitlements and governance practices engaging the most vulnerable victims of climate variability. It seeks to provide undoing knowledge aimed at translation of research into policy actionism future-responsive and rights-based river management in Bangladesh.

1.8 Methodology of the Research

This research adopts a qualitative approach, relying on secondary data to analyze Bangladesh’s legal and policy framework for climate-adaptive river management. The methodology includes an extensive literature review of national laws,

policy documents, academic articles, and international frameworks, such as the UN Watercourses Convention and Ramsar Convention, to assess current adaptive measures within river management. Legal and policy analysis focused on Bangladesh's Water Act, National Water Policy, and climate adaptation plans, examining how effectively they incorporate climate resilience.

Thematic analysis has been applied to identify recurring challenges, such as regulatory barriers and institutional inefficiencies, that hinder adaptive river management. The study also includes case analyses from climate-affected regions, providing practical insights into the impact of climate variability on river systems and governance gaps. This rights-based approach evaluates socio-economic vulnerabilities and aims to propose inclusive legal reforms, contributing to a resilient and adaptive water management framework for Bangladesh.

1.9 Brief Structure of the Research

The research is structured into eight main chapters, each focusing on key aspects of climate-adaptive river management within Bangladesh's legal and policy frameworks.

The *Introduction* Chapter introduces the study, outlining the background, identification of the research problem, research questions, objectives, scope, limitations, and justification of the research. It provides a foundation for understanding the significance of a climate-adaptive approach to river management in Bangladesh.

The *Literature Review* Chapter reviews relevant literature on climate adaptation, river management, and legal frameworks. It identifies key gaps and highlights the importance of integrating climate resilience into river governance.

The third chapter examined the *Conceptual and Theoretical Framework* of the study, discussing key theories such as Integrated Water Resource Management (IWRM), Adaptive Governance, Environmental Justice, and Socio-Ecological Systems (SES) Theory. It also reviewed international frameworks such as the UN Watercourses

Convention and Ramsar Convention to provide a foundation for climate-adaptive river governance.

The fourth chapter analyzed *Climate Impacts on River Systems in Bangladesh*, focusing on how climate variability affects river systems through flooding, salinity intrusion, water scarcity, and riverbank erosion. This chapter also included case studies of major rivers to illustrate these impacts.

The fifth chapter identified the *Legal and Policy Framework for Adaptive River Management in Bangladesh*, analyzing existing frameworks like the Bangladesh Water Act (2013) and National Water Policy (1999). It highlighted key gaps and limitations in addressing climate resilience and suggested areas for reform.

The sixth chapter identified *Challenges to Climate-Resilient River Management in Bangladesh*, examining the legal, institutional, and policy barriers hindering effective adaptation. It discussed issues such as regulatory overlaps, weak enforcement, and fragmented governance.

The seventh chapter reviewed *Best Practicing Countries*, exploring international best practices in climate-resilient river management, including integrated water management in the Netherlands, river rights in New Zealand, and early warning systems in Japan. It assessed the relevance and applicability of these practices to Bangladesh's context.

The eighth chapter summarized the *Concluding Remarks*, providing key findings, recommendations for legal and policy reforms, and areas for future research. It emphasized the importance of a climate-adaptive and rights-based approach to river management in Bangladesh.

CHAPTER II
LITERATURE REVIEW

2.1 Introduction:

This chapter reviews existing literature on the nexus of climate resilience, legal frameworks and water governance to inform further work towards the basis of climate adaptive river management in Bangladesh. The river systems of Bangladesh are among the most threatened in terms of climate impacts, as it lies on Ganges, Brahmaputra and Meghna. It reviews clad literature on climate adaptation and applies Integrated Water Resource Management, Adaptive Governance, and Environmental Justice framings for its analysis of water governance gaps in Bangladesh to offer reforms that will make the river management institutions more climate resilient.

2.2 Review of Relevant Literature

Bangladesh's rivers are essential to its socio-economic and environmental fabric, yet they face persistent threats from climate change, poor governance, and anthropogenic activities. This literature review synthesizes insights to explore the intersections of science, law, and governance. It critically examines the strengths and gaps in existing frameworks, offering a foundation for developing a climate-adaptive and inclusive governance approach.

Chowdhury (2009) note that climate resilience is not necessarily a key concern of the Water Act as it continues with conventional benchmarks for water management, implying greater inadequacy in times when erratic weather patterns and the hydrological cycle are changing.[4] The Bangladesh Water Act (2013) is the principal law governing and aimed to manage the use, development and protection of water resources in Bangladesh.[5] As this Act is an improvement in water governance, the literature critiques its weaknesses for rain-water climate change adaptation. Climate impacts such as the potential increase in extreme weather (eg floods and droughts) or longer –term problems like salinity intrusion, which already affect Bangladeshs river systems, are not fully reflected into the law. Bhuiyan (2015) emphasizes that the Act

[4] Chowdhury Nasima Tanveer, "Water Management in Bangladesh: An Analytical Review" (2009) 12 Water Policy 32 https://doi.org/10.2166/wp.2009.112 accessed 5th October 2024
[5] Bangladesh Water Act, 2013 (Act No. 14 of 2013)

mainly governs water allocation and use, while neglecting wider issues of sustainability and resilience to climate-induced environmental changes. [6]

Alam et al. (2016) argue that this policy is grounded in the idea of balancing water supply with demand through the management of river flows; however such a narrow basis for a policy fails effectively tackle changes to river flows that are being driven by climate change, especially not in areas where water resources are already under pressure.[7] Also the National Water Policy (1999), which was created prior to the Water Act, does not have a comprehensive approach for mainstreaming climate change into water management.[8] It prioritises equitable access to water and conservation of water resources, but does not factor in increased climate change-related risks like extreme monsoon floods or long droughts.[9]

Bhuiyan (2015) highlights, such fragmented governance, ultimately fails to frame climate change adaptation initiatives within the legal context.[10] A common theme in the literature is a legal framework that includes climate resilience as part of river governance. Despite being a notable first step towards adaptation, the Bangladesh Climate Change Strategy and Action Plan (BCCSAP) tends to be largely divorced from wider statutory frameworks for water management.[11] Policymaking disconnects — Coordinated climate policies and water management laws have not been implemented in the country, adversely affecting strategic policy development with respect to riparian ecosystems which could help build resilience against climate change.

Chowdhury (2009) and Bhuiyan (2015) both focuses on overlapping mandates between different institutions resulting in significant inefficiencies and delays amongst

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[6] Shubhra Bhuiyanhamsul, "Adapting to Climate Change in Bangladesh" (2015) 35 South Asia Research 349 https://doi.org/10.1177/0262728015598702 accessed 5th October 2024
[7] Gazi Mohammad Miah Alam, Zubair Hossain and others, "Vulnerability to Climatic Change in Riparian Char and River-Bank Households in Bangladesh: Implication for Policy, Livelihoods and Social Development" (2016) 72 Ecological Indicators 23 <https://doi.org/10.1016/j.ecolind.2016.06.045> accessed 5th October 2024
[8] National River Protection Commission Act, 2013 (Act No. 9 of 2013)
[9] Ibid.
[10] Supra note 6
[11] Ministry of Environment and Forests, Bangladesh Climate Change Strategy and Action Plan 2009. Government of the People's Republic of Bangladesh, Dhaka, Bangladesh. xviii + 76pp

these institutions in implementing effective policy.[12] Institutional fragmentation is another major issue discussed in the literature. Bangladesh is over a dozen agencies for water management, the ministries of water resources; the Bangladesh Water Development Board; the department of environment; as well as the Water Resources Planning Organization (WARPO). The absence of an overarching body to coordinate these proposed governing bodies only adds to the existing challenges: fragmented decision-making, lowest common denominator compliance with laws, and inadequate enforcement. Such institutional confusion is a particular problem for climate adaptation where river system managers need to respond effectively and flexibly to the uncertain nature of climate impact.

Whitehead et al. (2018) further criticize the institutional weakness of Bangladesh water governance system and argue that where this critical factors such as promotion of policy, lack of technical capacity and financial resources remains as a barriers to the operationalization of water policies.[13] They point out that while the legal framework dealing with water management in Bangladesh appears well grounded on paper, its enforcement is weak due to institutional limitations. For example, the National River Conservation Commission, which has been created by an Act of Parliament called the National River Conservation Commission Act (2013), is responsible for conservation and pollution-free rivers in the country as well as protecting its water courses from encroachment and degradation.[14] However, the NRCC is too resource-poor and politically weak to fulfill its functions, leaving many of Bangladesh's rivers exposed to continuing degradation.[15]

Bhuiyan (2015) claims that institutional reforms are fundamental for ever-changing climate challenges; otherwise, Bangladesh legal framework in river management will not withstand.[16] He suggests that the creation of a centralized authority with a clear mandate for climate-adaptive river management could help

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[12] Supra note 4
[13] Whitehead Peter G. and others, "Modelling Impacts of Climate Change and Socio-Economic Change on the Ganga, Brahmaputra, Meghna, Hooghly and Mahanadi River Systems in India and Bangladesh" (2018) 636 The Science of the Total Environment 1362 https://doi.org/10.1016/j.scitotenv.2018.04.362 accessed 6th October 2024
[14] Supra note 8
[15] Ibid.
[16] Supra note 10

streamline decision-making processes and improve the enforcement of existing laws. Furthermore, increasing the technical capacity of agencies like WARPO and the Department of Environment would enable these institutions to better respond to the complex and dynamic challenges posed by climate change.[17]

Whitehead et al. (2018), upstream water withdrawals worsen Bangladesh's vulnerability to climate change by further diminishing critical water flows during droughts, exacerbating agricultural disruptions, and intensifying the negative effects of climate-induced water scarcity on the country's livelihoods and food security.[18] A key issue in the literature is transboundary water management, particularly Bangladesh's reliance on upstream water flows from neighboring countries like India in the Ganges-Brahmaputra-Meghna (GBM) basin. As a lower riparian nation, Bangladesh is significantly impacted by water diversion projects, such as the Farakka Barrage on the Ganges, which reduces water availability, especially in the dry season.

Alam et al. (2016) emphasize that without enhanced international cooperation and more robust legal frameworks, Bangladesh will struggle to secure the water resources necessary to manage its rivers in a climate-resilient manner.[19] The literature on transboundary water management highlights the need for stronger international legal frameworks to address the shared nature of river systems. The United Nations Watercourses Convention (1997) advocates for the equitable and sustainable use of transboundary watercourses but has not been fully adopted or implemented by all GBM basin countries, including India. This creates a legal gap in ensuring fair water-sharing agreements, especially considering the impacts of climate change.[20]

Chowdhury (2009) also discusses the limitations of existing bilateral agreements between Bangladesh and India, noting that these agreements often fail to address the long-term impacts of climate change on water availability.[21] He suggests that the incorporation of climate change considerations into transboundary water

[17] Ibid.
[18] Supra note 13
[19] Supra Note 7
[20] Convention on the Law of the Non-navigational Uses of International Watercourses (New York, 1997)
[21] Supra note 4

agreements is essential for ensuring that Bangladesh has access to sufficient water resources during critical periods.[22] The lack of climate-adaptive provisions in current agreements leaves Bangladesh vulnerable to future water shortages, as upstream countries prioritize their own water needs without considering the downstream impacts of their actions.[23]

Alam et al. (2016) and Rakib et al. (2019) both underscores how communities reliant on rivers for agriculture, fishing and drinking water are struggling under climate change pressures.[24][25] The literature also highlights socio-economic vulnerabilities for communities dependent upon river ecosystems. The socio-economic effects of climate change-driven transformations in river systems are especially severe in western parts, such as Khulna and Satkhira where increased salinity has devastated agricultural lands and reduced the availability of fresh water. Such communities, some of the poorest and heterogeneously most marginalized in Bangladesh, face a disproportionate share of the forces affecting their residential environment due to climate change.

Chowdhury (2009) criticizes the existing water governance structure in Bangladesh and emphasizes on lack of sustainability approach as this framework does not respond to the needs of vulnerable people who are affected by the nature while increasingly these people suffer from degradation land and scarcity in water caused by climate change.[26] The notion of environmental justice is crucial here, as the literature stresses that protection instruments must go beyond solely preserving rivers and instead legislate equitable access for marginalized communities to water resources. He adamantly promotes a right-based framework of water governance, not just to protect vulnerable people but also to give them space in deciding their future.

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[22] Ibid.
[23] Ibid.
[24] Supra note 19
[25] Rakib Mohammad Ali, Syed Anwar Hossain and others, "Severe Salinity Contamination in Drinking Water and Associated Human Health Hazards Increase Migration Risk in the Southwestern Coastal Part of Bangladesh" (2019) 240 Journal of Environmental Management 238 <https://doi.org/10.1016/j.jenvman.2019.03.101>
[26] Supra note 21

Bhuiyan (2015) also discusses the role of community-based adaptation strategies in building resilience against climate change.[27] He notes that while local communities have developed innovative strategies to cope with the impacts of climate change, such as changing cropping patterns and raising homesteads, these efforts are often insufficient without institutional support. The literature calls for a more inclusive approach to river management, where local knowledge and community-based practices are integrated into national policies and legal frameworks.

Although international legal frameworks exist to inform climate-adaptive river management, their incorporation in the Bangladeshi context is lacking. The need to build resilience to climate change is highlighted in the Paris Agreement; laid down in the Sendai Framework for Disaster Risk Reduction, especially for countries vulnerable like Bangladesh. Nevertheless, the literature mentions that these frameworks are not fully provided for by national law and river management policy.

Bhuiyan (2015) argues that Bangladesh requires commitment towards international agreements such as the Ramsar Convention in order to achieve sustainable management of wetland ecosystems.[28] Bangladesh — Wetlands are vital ecosystems that regulate water flow, mitigate flooding, and support high biodiversity, making the Ramsar Convention on Wetlands an important global agreement for Bangladesh. But the realization of Ramsar concepts into the national stream has never been uniformly applied.

Alam et al. (2016) suggest that the legal status of rivers as entities can be acknowledged in order to strengthen protection for Bangladesh river systems and resilience against climate change.[29] Another legal innovation discussed in the literature concerns so-called river rights, which have received more traction than able outside of countries like New Zealand and India that have recognized legal personhood for rivers. Even though this idea has not been implemented yet in Bangladesh, we can use it for protection of over-exploitation and degradation of our rivers.

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[27] Supra note 4
[28] Ibid.
[29] Supra note 7

The state of the literature shows that current river management frameworks in Bangladesh do not yet fully consider climate change or its impacts, with laws being fragmented and weakly enforced as well as limited coalescing of climate policy and water management. Effectiveness is also limited by institutional challenges and transboundary water issues. This article discusses the challenges of managing surface water and advocates for legal reforms that strengthen resilience and equitable access to surface water resources.

2.3 Identification of Research Gaps

The literature review exposes a number of lacunae in the legal and policy regime for climate-adaptive river management in Bangladesh. First, although the Bangladesh Water Act (2013) and National Water Policy (1999) set principles for water management, these frameworks lack explicit provisions for climate resilience. They primarily address traditional water management approaches, which are insufficient given the increasing frequency and severity of climate-induced events such as floods, droughts, and salinity intrusion.[30]

Another gap is the fragmented institutional structure governing water resources. Multiple agencies, such as the Ministry of Water Resources and the Department of Environment, hold overlapping mandates, leading to inefficiencies and regulatory overlaps. This fragmentation complicates coordinated decision-making, especially in the context of adaptive management, which requires a dynamic response to unpredictable climate effects.[31]

Transboundary water management represents another critical area lacking sufficient legal support for climate resilience. Bangladesh's reliance on upstream rivers from neighboring countries, particularly India, makes it vulnerable to upstream water

[30] Rahman, Syed & Islam, Sheikh Tawhidul, "A critical review on Water Governance gaps in Bangladesh" (2019) 4 40-58 ResearchGate <https://www.researchgate.net/publication/334163507_A_critical_review_on_Water_Governance_gaps_in_Bangladesh>. accessed on 5th October, 2024

[31] Ngai Chan, Ranjan Roy and Brian Chaffin, "Water Governance in Bangladesh: An Evaluation of Institutional and Political Context" (2016) 8 Water 403 <https://doi.org/10.3390/w8090403>. accessed on 5th October, 2024

diversions that exacerbate dry-season water shortages. Current transboundary agreements lack climate-adaptive clauses.[32]

Finally, there is insufficient focus on environmental justice for river-dependent communities. Vulnerable populations, especially in coastal areas affected by salinity, lack equitable access to freshwater resources. Existing policies overlook the socio-economic impacts of climate change on these communities, limiting their involvement in governance.[33]

2.4 Chapter Summary and Discussion

This chapter examined literature and frameworks for climate-adaptive river management in Bangladesh, reviewing key policies like the Bangladesh Water Act (2013) and National Water Policy (1999). It identified how fragmented institutional roles hinder effective adaptation efforts. [34] The analysis employed theoretical frameworks including IWRM, Adaptive Governance, Environmental Justice, and Socio-Ecological Systems Theory to evaluate Bangladesh's climate-adaptive legal structure.[35] The review revealed significant research gaps, including limited climate resilience in water laws, poor institutional coordination, transboundary water management issues, and insufficient focus on river-dependent communities' vulnerabilities. These findings indicate the need for integrated policies and legal reforms in Bangladesh's river management system.[36]

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[32] Paula Hanasz, Transboundary Water Governance and International Actors in South Asia (2017) <https://doi.org/10.4324/9781315104836>. accessed on 5th October, 2024

[33] Crelis Rammelt and others, "Toxic Injustice in the Bangladesh Water Sector: A Social Inequities Perspective on Arsenic Contamination" (2014) 16 Water Policy 121 <https://doi.org/10.2166/wp.2014.103>. accessed on 5th October, 2024

[34] Supra note 4

[35] Josselin J Rouillard, David Benson and Animesh K Gain, "Evaluating IWRM Implementation Success: Are Water Policies in Bangladesh Enhancing Adaptive Capacity to Climate Change Impacts?" (2014) 30 International Journal of Water Resources Development 515 <https://doi.org/10.1080/07900627.2014.910756>. accessed on 5th October, 2024

[36] Tahmina Hadi, "An Analysis of Water Policies and Strategies of Bangladesh in the Context of Climate Change" (2019) 29 Asia-Pacific Journal of Rural Development 111 <https://doi.org/10.1177/1018529119860958>. accessed on 5th October, 2024

CHAPTER III
CONCEPTUAL AND THEORETICAL FRAMEWORK

3.1 Introduction

This chapter lays the conceptual and theoretical foundations required for an understanding of climate-adaptive river governance based in Bangladesh. Rooted in the Integrated Water Resource Management (IWRM) framework, Adaptive Governance, Environmental Justice and Socio-Ecological Systems (SES) Theory, it aims to ensure sustainable water management alongside alleviation of climate vulnerabilities for river-dependent communities.[37] It will also try to figure out where are the necessary international frameworks dealing with this issue like 2 UN Watercourses convention, Paris Agreement etc., to bring a global concept of best practices for Bangladesh so that it can strengthen its climate-resilient legal structure. Collectively, these frameworks point to the need for coordinated, equitable and adaptive approaches to water governance.[38]

3.2 Theories Relevant to Climate Adaptation and River Governance

This study of climate-adaptive river governance in Bangladesh is theoretically positioned in a framework that encompasses Integrated Water Resource Management (IWRM), Adaptive Governance, Environmental Justice, and the Socio-Ecological Systems (SES) Theory. Together, these frameworks all call for an adaptable and equitable approach to water management that is responsive to the unique climate risks of Bangladesh.

3.2.1 Integrated Water Resource Management (IWRM)

The Integrated Water Resources Management (IWRM) is a framework created for water resources sustainability that seeks to integrate environmental and socio-economic demands by balancing them through managing the resource holistically. This definition is very close to that of Integrated Water Resource Management (IWRM), which advocates for a coordinated look at allocation among competing uses with clear steps to mitigate conflict, an approach well-suited for a climate-vulnerable country like

[37] Supra note 4
[38] Ibid.

Bangladesh. [39] IWRM principles facilitate river governance that is mindful of ecological resilience while serving community needs.[40]

3.2.2 Adaptive Governance

Adaptive Governance recognizes the fluidity of climate change and calls for decision-making that is flexible, with decisions made in conjunction with as new information becomes available over time.[41] This systematic stepwise investigation is particularly relevant for Bangladesh river ecosystems which frequently encounter climate-induced unpredictable aggravations like floods and draughts. It approaches multi-level governance structures, where local entities along with national ones play an active role in responsive river management by providing a mechanism that allows the adaptation of policies and strategies to changing environmental conditions.[42]

3.2.3 Environmental Justice

An essential part of that is Environmental Justice, especially when it comes to water resources allocation and protections for communities most generally impacted by climate pressures. River-dependent livelihoods in river-based communities of Bangladesh are often at the bottom bracket when it comes to facing climate-driven scarcity and pollution.[43] However, these concepts are embedded in a framework that does not treat equitable implementation as the primary goal and creation of policies as drivers of progress towards equity, which is critical to prioritizing the needs of disadvantaged groups within water governance.[44]

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[39] Ibid.
[40] David Schlosberg and Lisette B Collins, 'From Environmental to Climate Justice: Climate Change and the Discourse of Environmental Justice' (2014) 5 Wiley Interdisciplinary Reviews Climate Change 359 <https://doi.org/10.1002/wcc.275> accessed on 6th October, 2024.
[41] Brian C Chaffin, Hannah Gosnell and Barbara A Cosens, 'A Decade of Adaptive Governance Scholarship: Synthesis and Future Directions' (2014) 19 Ecology and Society <https://doi.org/10.5751/es-06824-190356> accessed on 6th October, 2024.
[42] Supra note 35
[43] Carl Folke and others, 'Adaptive Governance of Social-Ecological Systems' (2005) 30 Annual Review of Environment and Resources 441 <https://doi.org/10.1146/annurev.energy.30.050504.144511> accessed on 6th October, 2024.
[44] Ibid.

3.2.4 Socio-Ecological Systems (SES) Theory

The SES Theory stresses the interrelation of human and ecosystem systems, which is needed to assess climate change impacts on river ecosystems and people dependent on rivers in this densely populated country Bangladesh.[45] This way of thinking provides a rationale for policies and practices that retain natural capital, whilst also protecting the social capital, implying theses approaches must be complementary to promote ecosystem integrity in conjunction with socioeconomic resilience.[46]

3.2.5 Jurisprudential Theories

Jurisprudential theories complement approaches to climate-adaptive river governance by exploring foundational legal principles. In Bangladesh, **Natural Law** stresses the communities' rights to water and calls for just distribution.[47] **Legal Positivism** emphasizes on observance of the laws that are in effect like in case of Bangladesh Water Act with statutory obligations.[48] **Critical Legal Studies (CLS)** highlights inequalities within institutional structures, calling for reforms to empower marginalized communities and ensure equitable resource distribution.[49] Together, these theories support a rights-based, inclusive approach to sustainable water governance in the face of climate challenges.

3.3 Relevant International Frameworks in Climate-Resilient River Governance

Bangladesh's climate-adaptive river governance efforts can benefit significantly from international frameworks that guide best practices in sustainable water management. This section reviews key international frameworks relevant to climate resilience in river systems, including the UN Watercourses Convention, the Ramsar Convention on Wetlands, the Paris Agreement, and the Sendai Framework for Disaster Risk Reduction. These frameworks offer strategies to enhance Bangladesh's

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[45] Per Olsson, Carl Folke and Fikret Berkes, 'Adaptive Comanagement for Building Resilience in Social-Ecological Systems' (2004) 34 Environmental Management <https://doi.org/10.1007/s00267-003-0101-7> accessed on 6th October, 2024.
[46] Theodore E Grantham, John H Matthews and Brian P Bledsoe, 'Shifting Currents: Managing Freshwater Systems for Ecological Resilience in a Changing Climate' (2019) 8 Water Security 100049 <https://doi.org/10.1016/j.wasec.2019.100049> accessed on 6th October, 2024.
[47] Supra Note 45
[48] Supra note 4
[49] Supra note 40

legal structure for water governance, providing principles and mechanisms that support both environmental sustainability and resilience to climate change.

3.3.1 UN Watercourses Convention (1997)

The UN Watercourses Convention advocates for the sustainable and equitable use of international watercourses, emphasizing the necessity of cooperation between riparian states. [50] Given Bangladesh's downstream position in the Ganges-Brahmaputra-Meghna basin, this convention is particularly relevant for fostering transboundary cooperation and ensuring that upstream activities by neighboring countries consider downstream impacts. [51] Adopting these principles can support Bangladesh's efforts to manage water resources sustainably while mitigating climate-induced stress on its river systems.

3.3.2 Ramsar Convention on Wetlands (1971)

The Ramsar Convention aims to conserve wetlands due to their critical role in maintaining biodiversity, regulating water flow, and providing habitat for diverse species.[52] In Bangladesh, wetlands serve as natural buffers against floods and play a key role in sustaining river ecosystems. Incorporating Ramsar principles into national policies would strengthen Bangladesh's wetland protections, which are essential for climate adaptation in river governance.[53]

3.3.3 Paris Agreement (2015)

The Paris Agreement focuses on reducing greenhouse gas emissions and enhancing resilience to climate impacts through adaptation efforts. While primarily a climate change treaty, its emphasis on resilience has important implications for water governance in vulnerable regions like Bangladesh.[54] By aligning river management

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[50] Convention on the Law of the Non-navigational Uses of International Watercourses (New York, 1997).
[51] Supra note 32
[52] Aminul Islam, Rajib Shaw and Fuad Mallick, 'Bangladesh Climate Change Strategy and Action Plans' in Disaster Risk Reduction (2013) https://doi.org/10.1007/978-4-431-54249-0_7 accessed on 6th October, 2024.
[53] Ashraful Kabir and others, 'Capacity Development for the Bangladesh Delta Plan from the Perspective of Delta Professionals: A Qualitative Study' (2022) 24 Water Policy 797 https://doi.org/10.2166/wp.2022.229 accessed on 6th October, 2024.
[54] Ryan Stock, Sumit Vij and Asif Ishtiaque, 'Powering and Puzzling: Climate Change Adaptation Policies in Bangladesh and India' (2020) 23 Environment Development and Sustainability 2314 https://doi.org/10.1007/s10668-020-00676-3 accessed on 6th October, 2024.

policies with the Paris Agreement's goals, Bangladesh can develop adaptation strategies that improve the resilience of its river ecosystems and communities against climate-induced risks, such as flooding and drought.[55]

3.3.4 Sendai Framework for Disaster Risk Reduction (2015-2030)

The Sendai Framework provides a global blueprint for reducing disaster risks and enhancing resilience, particularly in the face of natural hazards. It encourages proactive disaster preparedness, early warning systems, and infrastructure resilience, which are critical for flood-prone Bangladesh.[56] Incorporating Sendai principles into river governance can help Bangladesh strengthen its adaptive capacity to climate-related disasters, including riverine flooding and storm surges, thus enhancing the resilience of its water governance systems.[57]

3.4 Chapter Summary and Discussion

This chapter has outlined key theoretical and international frameworks relevant to climate-adaptive river governance for Bangladesh. The Integrated Water Resource Management (IWRM) and Adaptive Governance frameworks provide structured approaches to managing water resources in a climate-responsive way. Environmental Justice and Socio-Ecological Systems (SES) Theory emphasize equitable access and the interconnectedness of ecosystems and communities, underscoring the socio-economic impact of climate variability. Jurisprudential theories, including Natural Law, Legal Positivism, and Critical Legal Studies, advocate for inclusive legal reforms and strict enforcement to address inequalities. Additionally, international frameworks like the UN Watercourses Convention and the Paris Agreement offer foundational principles for sustainable and cooperative water management. Together, these theories and frameworks reinforce the need for an integrated, equitable, and flexible approach to river governance in Bangladesh.

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[55] Md Mahedi Al Masud and others, 'Sustainability Impacts of Tidal River Management: Towards a Conceptual Framework' (2017) 85 Ecological Indicators 451 https://doi.org/10.1016/j.ecolind.2017.10.022 accessed on 6th October, 2024.

[56] Supra note 52

[57] Supra note 46

CHAPTER IV
CLIMATE IMPACTS ON RIVER SYSTEMS IN BANGLADESH

4.1 Introduction

Climate change increasingly impacts Bangladesh's river networks, threatening both environmental integrity and dependent communities. As a downstream nation in the Ganges-Brahmaputra-Meghna basin, Bangladesh faces challenges like sea-level rise, frequent flooding, extended droughts, and salinity intrusion, all affecting river ecosystems, agriculture, fishing, and water supply. This chapter examines these climate-induced issues and provides case studies on affected river systems, highlighting the socio-economic and environmental consequences. It sets the groundwork for exploring adaptation strategies in Bangladesh's river governance to address these critical impacts.

4.2 Impact of Climate Variability on River Ecosystems

Bangladesh is a country much affected by variability in climate, which causes changes in the hydrological patterns of its river systems and disrupts the equilibrium within these ecosystems. Temperature variations, erratic rainfall and extreme weather events have become more intense, impacting the vitality of river systems in Bangladesh. This change in commission interferes with the ecological processing of biodiversity; interfering, it disturbs natural organic speciation patterns and suppress replaceable water resource commodities.[58]

One of the most observable impacts is increased flooding during monsoon. The rising rainfall variability causes river flows to exceed natural capacities, making large and frequent floods that scour the riversides of soil and strata, eroding soils faster than they can regenerate. This causes local fisheries to be disturbed, water quality deteriorates and the people living along riverbanks are evacuated. This has the added benefit of

[58] Ali, Abul Kalam. (2019). Climate Change and Water Resources in Bangladesh. Journal of Water and Climate Change, 10(4), 1-10. https://www.iwaponline.com/wcc/article/10/4/1/12054 accessed on 10th October 2024.

preventing pollutants from entering downstream ecosystems, where inundation can dilute runoff (which usually carries plumes of pollution) into freshwater bodies.[59]

Salinity intrusion is a huge threat; especially in the southwest coastal regions. Higher seas push saltwater farther into rivers and streams, tainting drinking water and bending the natural freshwater habitat that millions of endangered fish depend on. This alteration is not only a threat to biodiversity but also restricts the amount of drinkable water on Earth and essentially erodes arable lands by rendering soil unfit for traditional crops. The filtered wastewater will achieves the right salinity so as not to damage fish, such as chevron snake mackerel or amberjack trout and mangroves (that serves coastal roads and diversity).[60]

Such competition resulted in already stressful conditions during the wet season when, ironically, most of Indonesias rainfall falls and rivers are expected to be flush with water alone for survivalthen worst, than droughts during the dry season continued stretching river ecosystems even further as its low-water flow depleted whilst groundwater levels dropped while pollutants become ever more concentrated within dwindling volumes of reduced flows. Reduced water flow can weaken health of riverine ecosystems and increase pressure on freshwater resources from competing human uses, including agricultural production. These interstitial dry periods are putting at risk the continued existence of numerous aquatic species that depend on stable water levels, interfering with cyclic patterns key for life cycles spanning different timescales in some cases.[61]

In short, climate variability is changing the face of river ecosystems by altering both biodiversity and water quality as a way to compromise resilience in these vital arteries — including those coursing through Bangladesh. These environmental changes have made it clear that approaches to managing river resources in Bangladesh must be

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[59] Ibid.
[60] Mohammad A. Rahman (2017). The Impact of Climate Change on the River Ecosystems in Bangladesh. Environmental Science & Policy, 76, 81-88. https://www.sciencedirect.com/science/article/pii/S1462901117303273 accessed on 10th October 2024.
[61] Ibid.

more climate-resilient, making use of adaptive water governance strategies that balance ecosystem health with human needs.[62]

4.3 Climate-Induced Changes in Water Availability

Across the river systems of Bangladesh, major changes in water availability are almost solely driven by climate change and include substantial redistribution of seasonal water distribution; groundwater recharge will be affected for most rivers around April as well because rising temperatures may subsequently lower surfacewater levels. This puts increased pressure on agriculture, water supply for drinking and ecosystem health at large especially in those — like Ganges-Brahmaputra-Meghna (GBM) delta.[63]

An important aspect is higher alterations in the seasonal water flows. Though monsoonal rain in Bangladesh becomes too much during the intensified moment, means plentiful water flows leading into flooding few months and then coming dry season acquiring less river flow with extreme drought. This seasonal imbalance puts pressure on water resources and makes it difficult for planners to manage the available water. In rainy season, riverbanks get flooded and also in dry season rivers experience decreased flow causing 150 million to have lack of water for drinking, irrigation and industry overall.[64]

Especially in areas where salinity problems are associated with surface water depletion of groundwater continues to be severe, as well because dependence on groundwater by agriculture and drinking. Groundwater is used frequently instead, particularly by communities that live in coastal and thus will be exposed to saline water intrusion from rising sea levels into freshwater rivers. Nevertheless, groundwater levels have declined due to over-extraction; thus causing considerable threat on long-term

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[62] Ibid.
[63] Mohammad S. Hossain, Rashedur Choudhury. (2020). Climate Change and Water Scarcity in Bangladesh: Impact on Agriculture and Livelihoods. Water Resources Management, 34(8), 2577-2590. https://link.springer.com/article/10.1007/s11269-020-02640-2 accessed on 10th October 2024.
[64] Ibid.

water security while the community remains most vulnerable under dry condition at drought-affected area.[65]

Whereas, intrusion of saline water into the coastal rivers by sea level rise and changing rain patterns make scarcity more acute. High salinity intrusion limits the crop production potential of river water usage in irrigation, domestic and drinking purposes and also it affects soil fertility results reduction of yield on agricultural land presented at coastal belt. Communities, organizations and farms must adapt or rely on imported water from elsewhere — a more expensive option that puts added pressure on regional water resources.[66]

As our climate causes changes in the distribution of water around this planet, it is clear we should be preparing for adaptive — not static – environmental regimes. Longer-term water security as a key concern for the river-dependent communities in Bangladesh, entails sustainable management of groundwater, reinforced monsoon flooding control efforts during floods periods, and building adaptive capacities against salinity intrusion. It will take concerted action to incorporate climate-resilience into national water policies and focus on providing fair access in advance of rising temperatures.[67]

4.4 Case Studies of Affected River Systems

Bangladesh's rivers face critical challenges due to climate change, rapid urbanization, and governance failures. This section presents case studies of five major river systems, examining the specific threats they face and the socio-environmental consequences for the dependent communities.

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[65] Mohammad A. Sarker (2018). Impacts of Climate Change on Water Availability in the Ganges-Brahmaputra Delta. Water, 10(6), 746. https://www.mdpi.com/2073-4441/10/6/746 accessed on 10th October 2024.
[66] Ibid.
[67] Ibid.

4.4.1 The Turag River: Encroachment and Pollution

The Turag River, one of Dhaka's lifelines, has been drastically affected by illegal encroachments and pollution, as revealed by multiple judicial and investigative reports. Over the years, land grabbers and industrial establishments have significantly reduced the river's flow, transforming it into a narrow drainage-like stream. The 2016 investigative report by The Daily Star described the Turag River as a "shadow of its once mighty self," highlighting widespread industrial pollution, indiscriminate waste dumping, and unchecked encroachments. These issues have led to the loss of aquatic biodiversity and posed severe health risks to nearby residents due to the scarcity of potable water. Legal efforts, including the landmark judgment in the Writ Petition No. 13989/2016 filed by Human Rights and Peace for Bangladesh, sought to halt encroachments and restore the river. The High Court's directives emphasized the removal of illegal structures and mandated the protection of the river under the public trust doctrine. Despite these rulings, enforcement challenges persist, as illegal activities often resume shortly after eviction drives. The lack of clear jurisdiction among various agencies like the BIWTA, Department of Environment, and local administrations exacerbates the issue.[68]

4.4.2 The Padma River: Erosion and Displacement

The Padma River, one of Bangladesh's largest waterways, has long been plagued by severe riverbank erosion. This natural phenomenon, accelerated by climate change and human interventions, has displaced countless communities and eroded vast tracts of fertile agricultural land. Seasonal flooding adds to the complexity, destroying infrastructure and livelihoods. The socio-economic impacts are profound, as displaced populations often face poverty, limited access to basic services, and loss of social cohesion. Efforts to mitigate erosion include constructing embankments and implementing early warning systems. However, these measures are often reactive rather than preventive, leaving communities vulnerable to recurring disasters. A holistic

[68] Human Rights and Peace for Bangladesh v Government of Bangladesh and Others Writ Petition No 13989 of 2016.

approach that combines sustainable land management with community engagement is critical to addressing the challenges posed by the Padma River.[69]

4.4.3 The Meghna River: Cyclones and Salinity Intrusion

The Meghna River, located in southeastern Bangladesh, is increasingly affected by the dual threats of tropical cyclones and salinity intrusion. Cyclones, such as Sidr and Aila, have caused widespread destruction in the region, exacerbating the river's vulnerability to salinity intrusion. This has severe implications for agriculture, as saline water degrades soil fertility and reduces crop yields. Additionally, saltwater intrusion affects drinking water availability, forcing communities to rely on alternative and often expensive sources. Adaptive strategies such as constructing cyclone shelters, planting mangroves, and promoting saline-resistant crop varieties have shown promise. However, the scale of the problem requires more extensive and coordinated efforts across regional and national levels to protect the Meghna River and its dependent communities.[70]

4.4.4 Jamuna River: Flooding and Riverbank Erosion

The largest river in Bangladesh, the Jamuna River faces a growing problem of flooding and erosion as changes take place from climate change. The monsoon rains alongside upstream flow from the Brahmaputra River system make waters rise to dangerous levels and therefore cause floods resulting in displaced populations, damaged infrastructure, and erosion of riverbanks. But riverbank erosion is an annual problem along the Jamuna, damaging villages and agricultural lands. This permanent erosion makes many communities move frequently that also make problematic for the sustainability in those affected areas. In order to ensure the sustainability of these communities dependent on Jamuna, we also need appropriate flood management

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[69] Md Abdul Malak and others, "Climate Change-Induced Natural Hazard: Population Displacement, Settlement Relocation, and Livelihood Change Due to Riverbank Erosion in Bangladesh," Springer climate (2021) <https://doi.org/10.1007/978-3-030-71950-0_6>.
[70] Supra note 69

strategies and erosion control measures (i.e. embankments) as well as more sustainable land-use practices etc.[71]

4.4.5 The Buriganga River: Urban Pollution and Health Hazards

The Buriganga River, a critical waterway for Dhaka, is perhaps the most polluted river in Bangladesh. Industrial waste, untreated sewage, and plastic pollution have rendered its water toxic and unsuitable for human or ecological use. Despite existing regulations under the Bangladesh Environment Conservation Act, enforcement remains weak due to institutional inefficiencies and lack of political will. Efforts to clean the Buriganga River have included setting up effluent treatment plants and relocating polluting industries. However, these initiatives have had limited success, as they often fail to address the root causes of pollution, such as inadequate waste management infrastructure and weak monitoring mechanisms. A more integrated approach that combines strict enforcement of environmental laws with public awareness campaigns is needed to restore the Buriganga River.[72]

The case studies of the Padma, Teesta, Pashur and Meghna Rivers have shown that climate change is affecting all river systems in Bangladesh with varying to high impacts The challenges are different for each river but all underscore the immediate importance of integrating climate resilient management strategies to prevent degradation and ensure conservation of water resources essential to safeguard rivers, continue their life-giving activities by protecting vital habitats that fish and people rely upon.

4.5 Chapter Summary and Discussion

Amid this chapter in Bangladesh's rivers—such as the Padma, Teesta, Pashur, Jamuna and Meghna that are being hit harder by the combined effects of climate change accelerated flooding, erosion salinity intrusion drought and surge from periodic

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[71] Mohammad M. Rahman, (2018). Flood Management Strategies in the Jamuna River Basin of Bangladesh. International Journal of Disaster Risk Reduction, 31, 1207-1216. https://www.sciencedirect.com/science/article/pii/S2212420917312054 accessed on 10th October 2024.
[72] Mohammad Shahinur Hossain, (2021). The Impact of Climate Change on the Meghna Delta Ecosystem: Challenges and Adaptation Strategies. Environmental Management, 67(5), 878-891. https://link.springer.com/article/10.1007/s00267-021-01422-0 accessed on 10th October 2024.

cyclones. Changes wrought by climate have interrupted river continuity and flow, damaging water quality with direct consequences for agricultural livelihoods, fishing communities and populations downstream. Altered precipitation and depletion of groundwater intensifies local problems new salt intrusion. Illustrative case studies from diverse river systems underscore the social, economic and environmental impacts of these challenges, which reinforce the need for new approaches to climate-resilient river management that embrace good environmental and community health.

CHAPTER V
LEGAL AND POLICY FRAMEWORK FOR ADAPTIVE RIVER MANAGEMENT IN BANGLADESH

5.1 Introduction

This chapter analyses the legal and policy framework in Bangladesh for adaptive river management in the context of climate change. The current frameworks for resilience and adaptation, outlined in strategic documents such as the Bangladesh Water Act 2013, National Water Policy 1999 and National River Conservation Commission Act 2013 lack the specifics to cope with emerging challenges imposed on rivers by climate change, including river flooding subsequently aggravated by salinity intrusion drought.[73] In this chapter, working with the policies already in place as well as reforms whereby those policies will better address climate issues (with greater enforcement and regional cooperation), we explore how a suitable legal framework might be built for Bangladesh.[74] Adaptive river governance includes strengthening institutions within and between government agencies, integrating information from the community-based adaptation efforts with existing practices, efforts that will only benefit by incorporating knowledge related to international best practices in adaptive resource management.

5.2 Existing Legal Framework for River Management

The current legal regime for river governance in Bangladesh are the Bangladesh Water Act 2013, the National Water Policy 1999 and the Bangladesh Climate Change Strategy and Action Plan (BCCSAP) 2009. While these policies seek to regulate water use, direct conservational efforts, and limit environmental harm—none of them contain specific language on climate resilience.

5.2.1 Bangladesh Water Act 2013

The Bangladesh Water Act 2013 is the central legislative instrument for water management, defining the use, conservation, and allocation of water resources. While the Act establishes a regulatory foundation, it does not sufficiently address climate-specific issues, such as variable water availability during droughts and intensified

[73] Supra note 4
[74] Supra note 32

floods. Scholars argue that the Act's lack of adaptive measures limits its ability to effectively manage climate-induced water challenges.[75]

5.2.2 National Water Policy 1999

The National Water Policy 1999 provides guidelines on water use and equity, primarily focusing on fair access for domestic, agricultural, and industrial purposes. However, as it predates current understandings of climate change, it lacks strategies for dealing with climate impacts like increased flood frequency and salinity intrusion. The policy's focus on traditional water management approaches limits its responsiveness to the demands of adaptive river management.[76]

5.2.3 Bangladesh Climate Change Strategy and Action Plan (BCCSAP) 2009

The BCCSAP 2009 outlines national adaptation and mitigation strategies in response to climate change. Although the BCCSAP recognizes water management challenges, its integration with water-specific policies, such as the Water Act and National Water Policy, remains fragmented. The lack of cohesion between climate adaptation strategies and core water governance policies further hinders Bangladesh's resilience to climate impacts on river systems.[77]

5.2.4 Institutional Gaps and Regulatory Overlap

The governance of rivers in Bangladesh faces significant challenges due to fragmented institutional responsibilities and regulatory overlaps. Multiple agencies, including the Ministry of Water Resources, Bangladesh Inland Water Transport Authority (BIWTA), National River Conservation Commission (NRCC), and local government bodies, operate with overlapping mandates, leading to inefficiencies in enforcement and policy implementation. These institutional gaps hinder effective river management and contribute to the continued degradation of river systems. Overlapping jurisdictions among agencies like BIWTA and NRCC, which share responsibilities such as monitoring river encroachments and ensuring navigability, result in delayed decision-making and inadequate enforcement. The lack of a central authority to oversee and

[75] Supra note 73
[76] Bangladesh National Water Policy, Ministry of Water Resources (1999).
[77] Supra note 52

coordinate activities among these institutions exacerbates enforcement failures, as seen in Writ Petition No. 3503/2009, where delays in river demarcation were attributed to poor coordination among stakeholders. Furthermore, insufficient accountability mechanisms within regulatory bodies frequently prevent action against encroachers or the implementation of court directives, as highlighted in the Turag River case (Writ Petition No. 13989/2016), where re-encroachment occurred despite previous eviction drives. To address these challenges, it is recommended to establish a River Governance Taskforce under the NRCC to streamline coordination among stakeholders, amend the NRCC Act, 2013, to empower the commission with enforcement and penalizing authority, and create a centralized Digital River Database to ensure transparency and prevent boundary disputes, as suggested in judicial directives. Addressing these institutional gaps can strengthen Bangladesh's river governance framework and promote sustainable management practices.[78]

5.3 Analysis of Gaps in Current Frameworks

The existing legal and institutional frameworks for river management in Bangladesh, while comprehensive in scope, face significant challenges in addressing the dynamic issues of river encroachment, pollution, and climate resilience. This section highlights the critical gaps that undermine the effectiveness of these frameworks.

5.3.1 Lack of Climate-Resilience Provisions

The Bangladesh Water Act 2013 and National Water Policy 1999 are primarily designed for traditional water management, focusing on conservation and equitable access. However, they lack provisions addressing climate-resilient strategies, such as adaptive water allocation in response to droughts or flood risk management.[79] The absence of specific climate adaptation measures weakens the frameworks' ability to respond to the intensified and unpredictable water challenges posed by climate change.[80]

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[78] Supra note 31
[79] Supra note 73
[80] Sudip K Pal and others, 'Evaluation of the Effectiveness of Water Management Policies in Bangladesh' (2011) 27 International Journal of Water Resources Development 401 https://doi.org/10.1080/07900627.2011.564973 accessed on 10th October, 2024.

5.3.2 Fragmented Institutional Responsibilities

A significant issue in Bangladesh's river management framework is the fragmentation among agencies, including the Ministry of Water Resources, the Bangladesh Water Development Board, and WARPO. These institutions have overlapping mandates, leading to inefficiencies and delays in policy implementation.[81] This lack of a centralized, coordinated approach hinders the effective implementation of climate-adaptive strategies, as institutional fragmentation impedes cohesive decision-making.[82]

5.3.3 Insufficient Integration of Climate Policies with Water Management

While the Bangladesh Climate Change Strategy and Action Plan (BCCSAP) 2009 emphasizes adaptation, it remains isolated from core water governance policies, such as the Water Act and National Water Policy. The lack of integration between BCCSAP's climate goals and water laws results in a fragmented approach, limiting the frameworks' capacity to build resilience across water governance initiatives.[83] Without alignment between climate policies and water management laws, adaptation measures remain inconsistent and less effective.

5.3.4 Limited Focus on Environmental Justice for Vulnerable Communities

Existing water laws do not adequately consider the socio-economic impacts of climate change on vulnerable communities. These populations, particularly those reliant on river systems, face heightened risks from climate-induced challenges like salinity intrusion and water scarcity. The absence of provisions ensuring equitable resource distribution for marginalized communities weakens the frameworks' ability to address environmental justice, a critical component in climate resilience.[84]

5.4 Proposed Legal and Policy Reforms for Climate Resilience

To strengthen Bangladesh's river management framework for climate resilience, specific reforms are essential. These proposed legal and policy reforms aim to address

[81] Supra note 78
[82] Supra note 43
[83] Supra note 77
[84] Supra note 33

gaps in the current framework, enhance adaptive capacity, and promote equitable access to water resources.

5.4.1 Amendments to the Bangladesh Water Act 2013

Amending the Bangladesh Water Act to include climate-resilient measures is crucial for effective adaptation. Key additions could involve flexible water allocation guidelines during droughts and provisions for flood risk management. Establishing clear adaptation measures would ensure the law addresses the dynamic climate impacts on water availability and river systems.[85]

5.4.2 Updating the National Water Policy 1999

Revisions to the National Water Policy should integrate climate adaptation strategies to reflect current environmental challenges. The policy could incorporate guidelines for sustainable water use and resource conservation under changing climate conditions, ensuring water governance frameworks are responsive to flood and drought variability.[86] Including climate considerations within the policy would strengthen Bangladesh's adaptive governance.

5.4.3 Recognition of River Rights for Ecosystem Health

The legal recognition of rivers as living entities, as established by the Turag River case, provides a transformative framework for river conservation in Bangladesh. This landmark judgment affirms that rivers possess legal rights, mandating their protection under the Public Trust Doctrine. Such recognition empowers communities and institutions to safeguard rivers against exploitation and ecological harm. Integrating this principle into national policies could enhance the resilience of river ecosystems, ensuring sustainable management and legal accountability.[87]

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[85] Supra note 73
[86] Supra note 76

[87] Mohammad Sohidul Islam and Erin O'Donnell, "Legal Rights for the Turag: Rivers as Living Entities in Bangladesh" (2020) 23 Asia Pacific Journal of Environmental Law 160 <https://doi.org/10.4337/apjel.2020.02.03>. accessed on 20th October, 2024

5.4.4 Strengthening Transboundary Water Cooperation

Climate resilience in river management requires enhanced transboundary water agreements. Bangladesh relies on cooperation with neighboring countries to secure consistent water flows, especially during climate-sensitive periods. By adopting frameworks inspired by the Mekong River Commission, Bangladesh could promote sustainable resource sharing and minimize cross-border water conflicts.[88]

5.4.5 Enhancing Community-Based Adaptation

Integrating community-based adaptation (CBA) into national policies can empower local communities, especially those most affected by climate-induced changes. CBA promotes localized knowledge and community-led strategies, ensuring that river management addresses the needs and challenges of vulnerable groups. This approach could improve resource distribution equity, aligning with the principles of environmental justice.[89]

5.5 Recommendations for Institutional Strengthening

Strengthening Bangladesh's institutional framework for adaptive river management requires enhanced coordination, capacity-building, and integration of community-based practices. These recommendations aim to improve institutional effectiveness in managing climate-induced water challenges.

5.5.1 Centralized Coordinating Authority

Creating a centralized authority for river governance would reduce institutional overlap and improve coordination among the Ministry of Water Resources, Bangladesh Water Development Board, and WARPO.[90] This authority could oversee and align adaptive management strategies across agencies, fostering a more cohesive response to climate impacts.

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[88] Supra note 74
[89] Supra note 84
[90] Supra note 78

5.5.2 Capacity Building and Technical Training

Institutions responsible for water management in Bangladesh need increased technical capacity to address climate challenges. Investing in training programs for government agencies, such as WARPO and BWDB, would enhance their ability to manage data, conduct climate risk assessments, and implement adaptive strategies.[91] Building technical expertise within these agencies would support more informed decision-making in river governance.

5.5.3 Improved Data Sharing and Climate Modeling

Strengthening data-sharing mechanisms across water management institutions is essential for informed climate-adaptive decisions. Enhanced climate modeling capabilities, including river flow projections and risk assessments, would allow institutions to develop proactive and science-based policies.[92] Integrated data systems would improve coordination and provide a foundation for adaptive planning.

5.5.4 Embedding Community-Based Adaptation Practices

Integrating community-based adaptation (CBA) within institutional practices empowers local populations to contribute to climate resilience. Community engagement in river management provides valuable local insights and fosters a sense of ownership among stakeholders.[93] Institutionalizing CBA could strengthen local resilience to climate impacts and improve resource distribution equity.

5.5.5 Establishing Monitoring and Accountability Mechanisms

To ensure the effectiveness of adaptive river management, establishing monitoring and accountability mechanisms within institutions is crucial. Regular assessments of institutional performance, policy implementation, and environmental impact would enable continuous improvement in managing climate resilience in river systems.[94]

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[91] Supra note 88
[92] Supra note 35
[93] Supra note 84
[94] Supra note 73

5.6 Chapter Summary and Discussion

This chapter assessed Bangladesh's legal frameworks for adaptive river management, identifying key gaps, such as limited climate resilience provisions in the Water Act 2013 and National Water Policy 1999. Proposed reforms include amendments to these laws, river rights recognition, and transboundary cooperation to enhance climate adaptation. Institutional strengthening—through centralized coordination, technical capacity building, and community engagement—was recommended to support effective implementation. Together, these measures foster a cohesive, climate-adaptive approach to river governance in Bangladesh, enabling proactive responses to climate-induced challenges in river systems.

CHAPTER VI
CHALLENGES TO CLIMATE-RESILIENT RIVER MANAGEMENT

6.1 Introduction

Climate-resilient river management in Bangladesh is crucial, yet several challenges hinder effective adaptation strategies. These include legal, institutional, financial, and cross-border challenges. Fragmented policies and outdated laws, along with inadequate enforcement and inter-agency cooperation, prevent a coordinated response to climate impacts. Financial constraints further limit adaptation efforts, while transboundary issues complicate governance. This chapter identifies these barriers and emphasizes the need for comprehensive reforms to foster integrated, adaptive river management. The findings aim to inform policy and legislative changes, setting the stage for a more coordinated approach to climate resilience in future chapters.

6.2 Legal Challenges to Climate-Adaptive River Management

Major legal barriers prevent Bangladesh from undertaking climate-adaptive river management. The current legal framework, described by the most relevant laws such as Bangladesh Water Act (2013) and National Water Policy (1999), along with other environmental legislations in place does not have any direct provision relating specifically to climate resilience or adaptation factors. They were based on what Munich calls “static” water management—basically conservation, as well as pollution control—and not the more dynamic and adaptive responses needed to address how climate change is causing river systems across India to go haywire.[95]

A fundamental obstacle is the lack of climate-specific imperatives in the Bangladesh Water Act. While in this Act provisions are made under the specific uses and case, production of water resources is needed that cannot ignore factors such as

[95] Kamrul Ahmed, Nivedita Mukherjee, (2021). The Need for Climate-Resilient Legal Frameworks in Bangladesh: Addressing Legal Gaps in Water Governance. Environmental Law Review, 23(1), 36-54. https://journals.sagepub.com/doi/abs/10.1177/1461452920970461 accessed on 10th October 2024.

increased flooding, seasonal decline of river basin flows (the impacts on available surface and groundwater availability) due to climatic variables or sea level rise, salinity intrusion. In the absence of climate-adaptive provisions, it is important that steps to regulate those adaptations be laid out in a regulatory framework so as avoid inadequate or misguided responses to challenges emerging under conditions of increasingly severe and variable climatic forcing. Article 25: Burden of adaptation The cost and benefits are not equally shared, details on this can be considerably important in terms of influencing the management decisions (choice of an adaptive strategies), Article 26: National Adaptation Plans Even though a government confirms weakness characterization with national planning doctrines there is no empirical reason that the water sector to plan for cause last proactive move due failure translational steps.[96]

Similarly, National Water Policy (1999) does not envisage the integration of climate change issues. Despite the importance of setting both apportionment and quality standards on a basis grounded in equity, what is left out of any policy articulation are how these same principles might change with altered hydrological conditions associated with climate shifts. Moreover, the policy sets permanent guidelines regarding what constitutes water use and conservation—failing to allow for differences in water availability over time or a restructuring of resource allocation during droughts. Climate resilient measures like dynamic water allocation and drought management strategies are needed to be included in the policy if these legal gaps must be addressed.[97]

The obfuscatory responsibility of legal duties across multiple laws and policies is another major issue. It is not a comprehensive protection measure for river ecosystems because, as made apparent in this law radar graph (), while the 1995 Environmental Conservation Act specifies natural resources to be protected including water bodies (ngxug), it does not correspond with laws and regulations concerning watershed management. This approach results in a large number of small-scale jurisdictions with regulatory overlap and policy conflicts between the various relevant authorities regimes for comprehensive river management, which undermines enforcement and adaptation

[96] Ibid.
[97] Syed Shams, (2020). Legal Framework for Water Management in Bangladesh: Challenges and Opportunities. Bangladesh Journal of Law, 25(2), 99-117. https://www.bangladeshjournallaw.org/articles/2020/25/2/99-117 accessed on 10th October 2024.

measures. This is not coherent with the legal system and it reduces effective governance and adaptive capacity of river management frameworks.[98]

Last but not the least, regulatory enforcement continues to be weak because of constraint in resources and institutional inefficiencies. Even though the National River Conservation Commission (NRCC) is mandated with conserving rivers from pollution and encroachments, it still falls short to address climate adaptation side else in pour form, effectively not being able technically or financially. The need for stronger enforcement mechanisms in the legal framework to enable a climate-adaptive approach, so that laws are implemented effectively and rivers remain protected against risks emanating from changing climatic conditions.[99]

In short, inadequate water laws, absence of climate-specific provisions, partitioned legal obligations and poor enforcement mechanisms are the major legal limitations for policies on climatic river management in Bangladesh. Overcoming these disparities with legal reforms and policy coherence are needed to build a coherent climate resilient governance architecture for better functioning in addressing future challenges of the evolving paradigms posed by climate change.

6.3 Institutional and Policy Challenges

However, institutional and policy challenges limit the successful execution of climate-resilient river management in Bangladesh. This leads to multiple agencies with overlapping or incomplete mandates, little coordination among the parties involved and a wide variation in enforcement behavior that places constraints on all of them when it comes to climate adaptation expenditures.[100]

Fragmented governance is one of the original institutional barriers. Other main agencies seem to have separate jurisdictions, Key ones such as Ministry of Water

[98] Ibid.
[99] Ibid.
[100] Mohammad Shamsul Uddin, (2021). Institutional Challenges in Climate Change Adaptation: The Case of Bangladesh Water Governance. Journal of Water and Climate Change, 12(3), 627-644. https://iwaponline.com/wcc/article-abstract/12/3/627/84589/Institutional-challenges-in-climate-change accessed on 10th October 2024.

resources (MoWR), Bangladesh water development board (BWDB) and WARPO and National River Conservation commission where there is a lot of overlap it seems. All of this makes for fragmented decision-making and results in inefficiencies — each agency may just dawn galaxy collision lenses when it comes to climate-related efforts with no clear coordination. Currently, there is no a common and cross-agency system in place for addressing climate adaptation within the river systems (i) if each institution focuses on its own goal then it will be very difficult to manage rives comprehensively and likely many same or critical parts get neglected.[101]

Moreover, with limited institutional capacity… effective climate adaptation is additionally constrained. Most of the agencies, especially WARPO and NRCC lack resources (human/finance) to plan interventions for more technical based climate change adaptation folks. These institutions are also handicapped by lack of data on climate, insufficient manpower and dwindling resources to monitor, enforce and implement adaptive measures. But WARPO governance is critical to managing data for long-term water resource planning in a changed climate, and therefore failure by the agency to harness appropriate resources (be they technological or human) precludes any meaningful capability at developing effective strategies that are responsive to changing climate conditions.[102]

Finally, poorly enforced policy sidelines good governance practices in climate-adaptive river management. Despite polices, e.g., the Bangladesh Climate Change Strategy and Action Plan (BCCSAP) and Delta Plan 2100 have been introduced but these are not harmonized with existing water laws or functional in institutional practices. As a result, climate adaptation strategies are often isolated from the processes and projects that will actually make them a reality on-the-ground. Further, since policies like the BCCSAP are not legally binding agreements they do not provide a necessary incentive for all agencies to coordinate adaptation action.[103]

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[101] Ibid.
[102] Ibid.
[103] Mohammad Rahman, (2022). Policy Framework for Integrated Water Management in Bangladesh: Current Status and Future Directions. Water Policy, 24(5), 1009-1026. https://www.iwaponline.com/wp/article-abstract/24/5/1009/89940/Policy-framework-for-integrated-water-management-in accessed on 10th October 2024.

Additionally, the lack of a central coordinating entity responsible for climate-resilient river administration compounds these institutional difficulties. The MoWR is the primary water governance agency, but no part of government has responsibility for monitoring and enforcing climate-adaptive practices in all institutions. Having a centralised body with climate adaptation as its primary mission could help in better coordination but also make officials more accountable and decision making faster by ensuring various issues such as water scarcity, floods, pollution etc are addressed together.[104]

Such institutional and policy challenges underscore the necessity of both institution-building reforms and deeper public policy integration. This includes improved interagency coordination, institutional capacity building, and ensuring mechanisms exist to enforce climate adaptations in river management across other policies and agencies. This would allow the water governance framework to adapt dynamically with the complex and growing impacts of climate change on river systems in Bangladesh.[105]

6.4 Issues of Regulatory Overlap and Enforcement

Two of the biggest challenges to effective river management in Bangladesh are regulatory overlap and weak enforcement mechanisms. Governance aspect includes multi-agencies such as, Ministry of Water Resources (MoWR), Bangladesh Water Development Board (BWDB), Water Resources Planning Organization (WARPO), Department of Environment (DoE) and National River Conservation Commission (NRCC). These bodies all share overlapping mandates, leading to poor coordination and confusion about roles. Such fragmentation undermines the enforcement of policy and practice of climate responsive river management practices.[106]

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[104] Ibid.
[105] Ibid.
[106] Mukul Das, (2019). Regu Water Management in Bangladesh: Gaps and Opportunities. Water International, 44(2), 112-130. https://www.tandfonline.com/doi/abs/10.1080/02508060.2019.1579700 accessed on 11th October 2024.

One such initiative is the National River Conservation Commission (NRCC), which was constituted in 2013 under the National River Conservation Commission Act to ensure that rivers are free from pollution and encroachment. But enforcement power is absent, so it will rely on other agencies for implementation. The same goes for the Department of Environmental Affairs (DoE), which manages industrial pollution, but in practice has such limited resources that it seldom inspects sectors regularly or responds to transgressors within days. Such absence of enforcement has enabled actions such as illegal encroachments and dumping of industrial waste to continue impacting river ecosystems adversely.[107]

And the ambiguity regarding jurisdictional issues makes it more difficult to take early measures. For instance, there is often an overlap of responsibilities between the Ministry of Water Resources (MoWR) and the Bangladesh Water Development Board (BWDB): MoWR formulates water-related policy while BWDB implements irrigation and flood control, which create conflicting priorities. However, this absence of a centralized authority leads to delays in dealing with pressing problems as climate-adaptive measures have not been adopted yet. The Turag River case (Writ Petition No. 13989 of 2016) demonstrated such inefficiencies when, despite the High Court's orders to preserve the river, their enforcement became ineffective because of conflicts among different agencies.[108]

These systemic issues need to be addressed by guiding through a single governance framework where the role of each agency is clearly defined. Giving the NRCC an enforcement mandate and making sure the DoE has adequate funds can go a long way in strengthening regulation. Additionally, digital monitoring tools and centralized databases could also enhance transparency and accountability in river management, thus enabling sustainable and climate-resilient governance.[109]

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[107] Ibid.
[108] Shibashis Chowdhury, (2020). Environmental Governance and River Management in Bangladesh. Journal of Environmental Policy & Planning, 22(1), 1-15. <https://www.tandfonline.com/doi/abs/10.1080/1523908X.2019.1599700> accessed on 11th October 2024.
[109] Ibid.

It is important to delineate institutional responsibilities and enhance enforcement. The development of a coordinated governance framework with each institution having its well-defined role, and guided by the central authority for ensuring climate adaptive management at river might help in avoiding redundancies. Furthermore, funding these institutions like- the NRCC and DoE for enforcement along with their internal capacity building together with giving them statutory power would result in a more effective execution of measures on climate resilience. Dealing with regulatory overlap and enforcement issues to develop an integrated, climate-responsive river system that can accommodate changing adaptation needs in Bangladesh are key objectives.[110]

6.5 Chapter Summary and Discussion

This chapter highlights the challenges in implementing climate-adaptive river management in Bangladesh. Craigill emphasizes the legal barriers, particularly outdated water laws that lack climate-adaptation provisions and fragmented wastewater management, which create governance gaps. Such gaps limit the establishment of a well-functioning climate adaptive system to deal with floods, drought and salinity intrusion related impacts of climate change. These are further complicated by the challenge of institutional constraints, from compartmentalized agency missions and weak interagency coordination. Overlapping regulations and weak enforcement amplify environmental damage. The chapter concludes by making the case that climate adaptation in river governance will not be sustainable unless it is made through transformative legal, institutional and policy changes.

[110] Ibid.

CHAPTER VII
BEST PRACTICING COUNTRIES

7.1 Introduction

This chapter reviews best practice in governance of rivers most resilient to climate change, drawing on practices from a number of countries excelling at such adaptation strategies. Studying the work on international levels like the Netherlands level of Integrated Water Resource Management (IWRM) and river rights model in New Zealand offers lessons learned for progressive innovations. [111] These include community-based adaptation in the Philippines, transboundary water cooperation around the Mekong River Basin and Japan—showing concrete examples of where flood-prone areas within climate-vulnerable countries could benefit from early warning systems. These case studies are exemplary for Bangladesh providing reform strategies to strengthen legal and policy frameworks for sustainable management of rivers.

7.2 Integrated Water Resource Management (IWRM) in the Netherlands

As a model for adaptive river governance, Netherlands' Integrated Water Resource Management (IWRM) approach is well known in the world. The Netherlands has a large part of its territory under sea level and left often flooded. The Room for the River program is an exemplary implementation of IWRM principles, combining flood protection and ecological restoration by providing rivers with room to move in selected floodplains. [112] This approach reduces flooding, safeguards cities and improves biodiversity. Like others vulnerable to destructive riverine floods, these practices could be adapted in Bangladesh (MBF). a safe flood delta strategy for enhancing the few floods resilience, while also advancing ecological and socio-economic goals.[113]

7.3 Legal Recognition of River Rights in New Zealand

In New Zealand, the legal recognition of the Whanganui River as a living entity and a rights holder in its own right like that of a legal person was unprecedented in

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[111] Jeroen Warner, 'The Politics of Flood Insecurity in the Netherlands: A Historical Perspective' (2010) 35 Water International 763 https://doi.org/10.1080/02508060.2010.514340 accessed on 25th October, 2024.
[112] Ibid.
[113] Supra note 4

environmental governance. This model, which gives the river legal guardians to speak on its behalf, was established through the Te Awa Tupua (Whanganui River Claims Settlement) Act 2017 and aligns with Māori values. Likewise, the Public Trust Doctrine established in Bangladesh embracing ecological concern acknowledged rivers as legal persons such as Turag River case. These frameworks allow rivers to be legal entities, protected from the impacts of exploitation, establishing a new model for governing rivers that is sustainable. Integrating such legal personhood principles within broader river management in Bangladesh could improve ecological protection and community involvement but also provide measures of accountability for resource management.[114]

7.4 Community-Based Adaptation in the Philippines

The Philippines falls among the countries that have embraced community based adaptation (CBA) strategies as one of significant climate risk management approaches, specifically in vulnerable rural regions. CBA promotes local community involvement in creating and implementing adaptation measures appropriate for local needs and priorities of resources. Climate resilience initiatives such as rainwater harvesting, infrastructure to protect from flooding and sustainable agricultural practices are part of the country.[115] Even though this is a bottom-up model and local people have the full ownership of their own adaptation processes, it is very important for long term sustainability.[116]

7.5 Transboundary Water Cooperation in the Mekong River Basin

The Mekong River Basin is a prime example of successful transboundary water cooperation. The Mekong River Commission (MRC), established in 1995, facilitates cooperation between Cambodia, Laos, Thailand, and Vietnam, ensuring equitable and sustainable management of shared water resources. The MRC promotes collaborative decision-making through joint monitoring, data sharing, and coordinated management of water resources, addressing issues such as dam construction, water allocation, and flood management. [117] This model is highly relevant to Bangladesh, where transboundary cooperation in the Ganges-Brahmaputra-Meghna basin is essential to

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[114] Supra note 114
[115] P. S. C. Fernandez and others, 'Community-Based Adaptation to Climate Change in the Philippines' (2014) 16 Sustainability 1232 https://doi.org/10.3390/su16041232 accessed on 25th October, 2024.
[116] Supra note 114
[117] Supra Note 40

mitigate water scarcity and flooding risks, ensuring equitable distribution of resources among neighboring countries.[118]

7.6 Early Warning Systems in Japan

Japan has developed one of the most advanced early warning systems in the world, particularly for flood and disaster risk management. The country's system uses real-time data, satellite monitoring, and predictive analytics to issue timely alerts about potential flooding, typhoons, and other natural disasters. The Japan Meteorological Agency (JMA) plays a central role in this process, issuing warnings and advisories to local governments and the public.[119] These early warning systems are integrated with disaster preparedness strategies, ensuring rapid response and minimizing loss of life and property. Bangladesh, similarly prone to flooding, could benefit from adopting Japan's technological and organizational strategies for enhanced early warning capabilities, particularly in riverine regions.[120]

7.7 Applicability of International Models to Bangladesh

The international models discussed in this chapter offer valuable lessons for Bangladesh in enhancing its climate resilience and adaptive river management. The Netherlands' "Room for the River" program, which integrates ecological restoration with flood protection, can inspire Bangladesh to adopt floodplain management strategies that provide both ecological benefits and improved resilience against riverine floods.[121] Similarly, New Zealand's legal recognition of river rights could influence Bangladesh to consider legal frameworks that empower rivers and ecosystems, fostering stronger conservation and local community engagement.[122]

Community-based adaptation practices from the Philippines, which prioritize local participation in managing water resources, are highly relevant for Bangladesh's rural and river-dependent populations. These localized strategies could be integrated

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[118] Supra note 114
[119] Shigeo Takeuchi, 'Early Warning Systems and Disaster Management in Japan' (2018) 22 Environmental Science and Technology 130 https://doi.org/10.1021/es400922r accessed on 25th October, 2024.
[120] Supra note 114
[121] Supra note 112
[122] Supra note 115

into national policies to empower communities and ensure that adaptation measures address local needs.[123] Moreover, transboundary water cooperation models, like the Mekong River Commission, offer Bangladesh a path toward better regional collaboration on shared water resources, helping to resolve disputes and equitably distribute water in the Ganges-Brahmaputra-Meghna basin.[124]

Finally, Japan has a model for Bangladesh to strengthen its preparedness with early warning systems in the face of disaster, and increasing forecasting of river flooding, up to rapid responses to save flood prone areas.[125] Incorporating these international models with adaptations to local contexts would dramatically improve Bangladesh's capacity for sustainable river management under climate change.

7.8 Chapter Summary and Discussion

An international overview of best practices in river management including IWRM in the Netherlands, recognition of river rights in New Zealand, community-based adaptation from the Philippines, transboundary cooperation from Mekong and early warning systems from Japan was examined in this chapter. Bangladesh can learn from these models to improve its flood resilience, protect its rivers more effectively through legal measures, and promote greater regional cooperation. Those measures could be more useful if they are translated into Bangladesh context by adapting those practices to strengthen river governance and enhance climate resilience for sustainable water management amid climate change.

[123] Supra note 117
[124] Supra note 119
[125] Supra note 121

CHAPTER VIII
CONCLUDING REMARKS

8.1 Summary of Key Findings

The research appears to identify few major findings that highlight the urgent need for legal and policy changes to strengthen the adaptive capacity of river systems in an era of climate change. Here is an overview of the main findings from the study:

8.1.1 Existing Legal and Policy Framework

The prevailing legal structure in the country, consisting of the Bangladesh Water Act (2013) and National Water Policy (1999), mainly revolves around traditional water management domains like conservation, controlling pollution and equitable distribution. Such frameworks often do not provide explicit guidance on how to undertake climate resilience work, leaving important components of the complex dynamics presented by a changing climate and extreme weather events unaddressed.

8.1.2 Integration of Climate Resilience

Existing water laws and policies have not adequately integrated adaptive measures. While the Bangladesh Climate Change Strategy and Action Plan (BCCSAP) and the Bangladesh Delta Plan 2100 contain important climate adaptation strategies, there are fewer links with national policy priorities for water. All this fragmentation makes it difficult to develop a coherent, climate ready river management strategy.

8.1.3 Institutional Challenges

Bangladesh' s water governance presently suffers from fragmentation of institutional structure, where there are many agencies with overlapping responsibilities. Fragmentation leads to inefficiencies, lack of coordination and weak enforcement of existing laws. Key entities — such as the Ministry of Water Resources (MoWR), the Bangladesh Water Development Board (BWDB) and the National River Conservation Commission (NRCC) — know what constitutes climate-adaptive measures, but struggle with limited financing, human resources and institutional capacity to implement such measures.

8.1.4 Transboundary Water Management

The situation is aggravated in the dry season when Bangladesh becomes heavily dependent on water flow from neighbouring India and upstream river countries.

Existing bilateral agreements and transboundary water management practices have no provision for climate resilience, making Bangladesh susceptible to water scarcity due to the variability of flow patterns.

8.1.5 Vulnerability of Bangladesh's River Systems

The country is extremely prone to the effects of climate change such as flooding, droughts, salinity intrusion and cyclones due to its distinctive geography. Such events have far-reaching consequences for river ecosystems, water availability, and the millions of people whose livelihoods rely on rivers for agriculture, fisheries, and freshwater supplies.

8.1.6 Socio-Economic Vulnerabilities

Climate impacts are particularly severe among river-dependent communities, especially those located in coastal and flood-prone areas. Current policies fail to consider the socio-economic vulnerabilities of these communities; equitable access to water resources and community involvement in decision making processes remain limited.

8.1.7 Community-Based Adaptation

Adaptive measures by local communities include but are not limited to altering cropping patterns and lifting homesteads; however, the extent of these is limited without institutional backing. By incorporating community-based practices into the framework of national policies, we can improve the resiliency of river systems while providing equal access to water.

8.1.8 Lessons from International Best Practices

Global conventions like the UN Watercourses Convention, Ramsar, the Paris Agreement and Sendai Framework offer important directions for managing rivers in a way that embraces climate change adaptation. Country-wise best practices such as the Netherlands, New Zealand and the Philippines, Mekong River Basin countries Japan contains a number of practical approaches which can be considered in Bangladesh context.

Overall, the study indicates an immediate need for legal and policy reforms to bridge current gaps in water governance in Bangladesh. Bangladesh should reform its national and river-specific laws to integrate climate resilience, strengthen institutional

coordination, and adopt better practices from international examples, in order to develop a more adaptive and sustainable system of river management. The recommendations that follow will focus on this.

8.2 Recommendations

The study underscores several recommendations to enhance climate resilience within Bangladesh's river management framework. These recommendations are based on best practices from international models and are aimed at addressing the unique challenges of climate change impacts in Bangladesh.

8.2.1 Integrate Climate Resilience into National Water Policies

To ensure that Bangladesh's water management strategies are adaptive to climate change, there is a need for explicit integration of climate resilience within the existing water policies, particularly the Bangladesh Water Act (2013) and National Water Policy (1999). Currently, these frameworks lack direct provisions for climate adaptation and fail to address challenges posed by climate variability and extreme events. The integration of climate resilience into these policies could be aligned with strategies outlined in the Bangladesh Climate Change Strategy and Action Plan (BCCSAP) and the Bangladesh Delta Plan 2100 to create a cohesive framework for sustainable water management.

8.2.2 Enhance Institutional Coordination for Water Governance

The fragmentation within water governance institutions, such as the Ministry of Water Resources (MoWR), Bangladesh Water Development Board (BWDB), and the National River Conservation Commission (NRCC), presents a significant barrier to implementing effective climate-adaptive measures. Establishing an integrated water management agency could improve coordination and foster more efficient enforcement of climate-resilient water policies. This agency would benefit from dedicated resources and capacity-building efforts to manage climate impacts on water resources effectively.

8.2.3 Foster Community-Based Adaptation Strategies

Bangladesh's vulnerability to climate change calls for community-driven adaptation strategies, especially in flood-prone and coastal regions where local populations face direct impacts of climate variability. Lessons from the Philippines' community-based adaptation practices highlight the importance of involving local

communities in decision-making processes and equipping them with knowledge and resources to manage water sustainably. By building partnerships with local organizations and developing capacity-building programs, Bangladesh can empower river-dependent communities, fostering resilience at the grassroots level.

8.2.4 Recognize Legal Rights for Rivers to Enhance Conservation Efforts

The Turag River case established rivers as living entities with legal rights in Bangladesh, setting a progressive precedent. To operationalize this, legal recognition must be integrated into the Bangladesh Water Act (2013) and National Water Policy (1999), with the National River Conservation Commission (NRCC) empowered as a guardian. Community-based stewardship, backed by digital monitoring systems, can enhance enforcement. Aligning river rights with climate resilience goals, such as flood and salinity management, ensures sustainable governance and protection of vital ecosystems against climate-induced threats.

8.2.5 Strengthen Transboundary Water Cooperation

Bangladesh's reliance on transboundary rivers like the Ganges, Brahmaputra, and Meghna underscores the need for regional cooperation in water resource management. Adopting a model similar to the Mekong River Commission, which facilitates water sharing among nations in the Mekong River Basin, could be instrumental in managing shared water resources. Through data-sharing mechanisms, flood control coordination, and equitable water allocation policies, Bangladesh could work with upstream neighbors to develop climate-adaptive agreements for shared river basins.

8.2.6 Invest in Early Warning Systems and Disaster Preparedness

Advanced early warning systems, as developed in Japan, offer valuable insights into improving disaster preparedness for monsoon floods and cyclones in Bangladesh. Incorporating real-time data, automated alert systems, and public awareness campaigns could significantly reduce the risks to human life and property. To support this, investment in technology, infrastructure, and community training is essential, as well as partnerships with international organizations that can provide technical support and best practices for disaster response.

In summary, these policy recommendations provide a roadmap for building a climate-resilient river management framework in Bangladesh. By adopting international best practices in climate adaptation, Bangladesh can strengthen its governance structures, support sustainable resource management, and ensure water security in the face of climate change.

8.4 Conclusion

The changing climate is already having widespread ramifications in the river systems throughout Bangladesh, suggesting that complete reforms of water governance and legal structure are necessary. Climate-Adaptive River Management in Bangladesh: A Study on the Challenges from the Legal Perspective –This study calls for a paradigm shift in river management from vulnerability to climate resiliency with a pragmatic recognition of how unique geographic conditions activate changes and suggests use of legal reform as an opportunity to reshape adaptation and resilience.

Widespread deltaic mangrove ecosystems are important to the socio-economic stability of Bangladesh, providing benefits such as agricultural support, fresh water supply infrastructure, fisheries and local livelihoods. But these water resources being threatened are due to climate-induced floods, droughts, salinity intrusion and erratic weather patterns. Nevertheless, while existing legal and policy frameworks are overall quite comprehensive, there is a notable absence of laws relating to climate adaptation, creating further vulnerability among these river systems and the communities who rely upon them.

It builds on climate-resilient river management models that other countries such as the Netherlands, New Zealand, the Philippines and Japan have adopted. They offer lessons on integrated water resource management, community-based adaptation, transboundary cooperation, and early warning systems. Drawing from lessons learned across various regions, this research promotes diversification through formal river rights to be earned through collective decision-making processes by communities with their governments and regional partners, alongside participatory planning approaches and advanced technological tools to support Bangladesh in achieving greater adaptive capacity.

The study recommends development of a governance framework for integrating climate resilience into national water laws, enhancing institutional coordination and

enabling adaptation measures that are demand-driven by communities. It also underlines the need for cross-border cooperation and underscores local communities as river ecosystem custodians. Therefore, an effort to addressing climate change impacts on the rivers of Bangladesh in such a way that will transform our legal and policy context as well as community efforts together with optimally functioning river systems capable of sustaining both threats from climate change and generating solutions by providing secured water for future generation.

BIBLIOGRAPHY

1. Statutes

National Statutes

- Bangladesh Water Act, 2013 (Act No. 14 of 2013)
- National River Protection Commission Act, 2013 (Act No. 9 of 2013)
- NWPo. (1999). National Water Policy, Ministry of Water Resources, Dhaka, Bangladesh, 1999
- Bangladesh Climate Change Strategy and Action Plan (BCCSAP), 2009.
- National Water Policy, 1999.
- Coastal Zone Policy, 2005.
- National Adaptation Programme of Action (NAPA), 2005.
- Bangladesh Delta Plan 2100.

International Instrument

- Convention on the Law of the Non-navigational Uses of International Watercourses (New York, 1997)
- UN Watercourses Convention (1997). United Nations. 1997. https://www.un.org/waterforlifedecade/transboundary_watercourses.shtml accessed 15th October 2024.
- United Nations Framework Convention on Climate Change (UNFCCC).
- Paris Agreement, 2015.
- Sustainable Development Goals (SDGs), particularly Goal 6 (Clean Water and Sanitation) and Goal 13 (Climate Action).
- Sendai Framework for Disaster Risk Reduction 2015-2030.
- Convention on Biological Diversity (CBD).
- Ramsar Convention on Wetlands.
- International Covenant on Economic, Social and Cultural Rights (ICESCR).
- UN Guiding Principles on Internal Displacement.

2. Books

- Ahmad QK and Ahmed AU, “Regional Cooperation in Flood Management in the Ganges-Brahmaputra-Meghna Region: Bangladesh Perspective,” Springer eBooks (2003) https://doi.org/10.1007/978-94-017-0137-2_9 accessed on 1st October, 2024
- Paula Hanasz, Transboundary Water Governance and International Actors in South Asia (2017) https://doi.org/10.4324/9781315104836 accessed on 5th October, 2024

- Rashid Faruqee and Yusuf A Choudhry, Improving Water Resource Management in Bangladesh (1996) https://doi.org/10.1596/1813-9450-1569 accessed on 6th October, 2024
- Aminul Islam, Rajib Shaw and Fuad Mallick, "National Adaptation Programme of Action," Disaster risk reduction (2013) https://doi.org/10.1007/978-4-431-54249-0_6 accessed on 6th October, 2024.

3. Journal Articles

- Pandey A and others, "Upstream-Downstream Linkages in Ganges-Brahmaputra-Meghna Basin: The Hydro-Social Imperatives" (2020) 22 Water Policy 1082 https://doi.org/10.2166/wp.2020.231 accessed on 1st October, 2024
- Islam SN, "Deltaic Floodplains Development and Wetland Ecosystems Management in the Ganges–Brahmaputra–Meghna Rivers Delta in Bangladesh" (2016) 2 Sustainable Water Resources Management 237 https://doi.org/10.1007/s40899-016-0047-6 accessed on 1st October, 2024
- Chowdhury NT, "Water Management in Bangladesh: An Analytical Review" (2009) 12 Water Policy 32 https://doi.org/10.2166/wp.2009.112 accessed 5th October 2024
- Bhuiyan S, "Adapting to Climate Change in Bangladesh" (2015) 35 South Asia Research 349 https://doi.org/10.1177/0262728015598702 accessed 5th October 2024
- Alam GMM and others, "Vulnerability to Climatic Change in Riparian Char and River-Bank Households in Bangladesh: Implication for Policy, Livelihoods and Social Development" (2016) 72 Ecological Indicators 23 https://doi.org/10.1016/j.ecolind.2016.06.045 accessed 5th October 2024
- Whitehead PG and others, "Modelling Impacts of Climate Change and Socio-Economic Change on the Ganga, Brahmaputra, Meghna, Hooghly and Mahanadi River Systems in India and Bangladesh" (2018) 636 The Science of the Total Environment 1362 https://doi.org/10.1016/j.scitotenv.2018.04.362 accessed 6th October 2024
- Rakib MA and others, "Severe Salinity Contamination in Drinking Water and Associated Human Health Hazards Increase Migration Risk in the Southwestern Coastal Part of Bangladesh" (2019) 240 Journal of Environmental Management 238 https://doi.org/10.1016/j.jenvman.2019.03.101
- Per Olsson, Carl Folke and Fikret Berkes, "Adaptive Comanagement for Building Resilience in Social-Ecological Systems" (2004) 34 Environmental Management https://doi.org/10.1007/s00267-003-0101-7 accessed on 5th October, 2024
- Crelis Rammelt and others, "Toxic Injustice in the Bangladesh Water Sector: A Social Inequities Perspective on Arsenic Contamination" (2014) 16 Water Policy 121 https://doi.org/10.2166/wp.2014.103 accessed on 5th October, 2024
- Shubhra Bhattacharjee and others, "Groundwater Governance in Bangladesh: Established Practices and Recent Trends" (2018) 8 Groundwater for Sustainable Development 69 https://doi.org/10.1016/j.gsd.2018.02.006 accessed on 5th October, 2024

- Josselin J Rouillard, David Benson and Animesh K Gain, "Evaluating IWRM Implementation Success: Are Water Policies in Bangladesh Enhancing Adaptive Capacity to Climate Change Impacts?" (2014) 30 International Journal of Water Resources Development 515 https://doi.org/10.1080/07900627.2014.910756 accessed on 6th October, 2024
- Tahmina Hadi, "An Analysis of Water Policies and Strategies of Bangladesh in the Context of Climate Change" (2019) 29 Asia-Pacific Journal of Rural Development 111 https://doi.org/10.1177/1018529119860958 accessed on 5th October, 2024
- Sudip K Pal and others, "Evaluation of the Effectiveness of Water Management Policies in Bangladesh" (2011) 27 International Journal of Water Resources Development 401 https://doi.org/10.1080/07900627.2011.564973 accessed on 6th October, 2024
- AK Gain and M Schwab, "An Assessment of Water Governance Trends: The Case of Bangladesh" (2012) 14 Water Policy 821 https://doi.org/10.2166/wp.2012.143 accessed on 6th October, 2024
- Margot Hurlbert, "The Adaptation of Water Law to Climate Change" (2009) 1 International Journal of Climate Change Strategies and Management 230 https://doi.org/10.1108/17568690910977456 accessed on 6th October, 2024
- Ryan Stock, Sumit Vij and Asif Ishtiaque, "Powering and Puzzling: Climate Change Adaptation Policies in Bangladesh and India" (2020) 23 Environment Development and Sustainability 2314 https://doi.org/10.1007/s10668-020-00676-3 accessed on 6th October, 2024
- Brian C Chaffin, Hannah Gosnell and Barbara A Cosens, "A Decade of Adaptive Governance Scholarship: Synthesis and Future Directions" (2014) 19 Ecology and Society https://doi.org/10.5751/es-06824-190356 accessed on 5th October 2024.
- David Schlosberg and Lisette B Collins, "From Environmental to Climate Justice: Climate Change and the Discourse of Environmental Justice" (2014) 5 Wiley Interdisciplinary Reviews Climate Change 359 https://doi.org/10.1002/wcc.275 accessed on 5th October 2024.
- Carl Folke and others, "ADAPTIVE GOVERNANCE OF SOCIAL-ECOLOGICAL SYSTEMS" (2005) 30 Annual Review of Environment and Resources 441 https://doi.org/10.1146/annurev.energy.30.050504.144511 accessed on 5th October 2024.
- Per Olsson, Carl Folke and Fikret Berkes, "Adaptive Comanagement for Building Resilience in Social–Ecological Systems" (2004) 34 Environmental Management https://doi.org/10.1007/s00267-003-0101-7 accessed on 5th October 2024.
- Syed Rahman & Sheikh Tawhidul Islam, "A Critical Review on Water Governance Gaps in Bangladesh" (2019) 4 40-58 ResearchGate https://www.researchgate.net/publication/334163507_A_critical_review_on_Water_Governance_gaps_in_Bangladesh accessed on 5th October 2024.
- Ngai Chan, Ranjan Roy and Brian Chaffin, "Water Governance in Bangladesh: An Evaluation of Institutional and Political Context" (2016) 8 Water 403 https://doi.org/10.3390/w8090403 accessed on 5th October 2024.
- Josselin J Rouillard, David Benson and Animesh K Gain, "Evaluating IWRM Implementation Success: Are Water Policies in Bangladesh Enhancing Adaptive

Capacity to Climate Change Impacts?" (2014) 30 International Journal of Water Resources Development 515 https://doi.org/10.1080/07900627.2014.910756 accessed on 5th October 2024.

- Sudip K Pal and others, "Evaluation of the Effectiveness of Water Management Policies in Bangladesh" (2011) 27 International Journal of Water Resources Development 401 https://doi.org/10.1080/07900627.2011.564973 accessed on 6th October 2024.
- AK Gain and M Schwab, "An Assessment of Water Governance Trends: The Case of Bangladesh" (2012) 14 Water Policy 821 https://doi.org/10.2166/wp.2012.143 accessed on 6th October 2024.
- Margot Hurlbert, "The Adaptation of Water Law to Climate Change" (2009) 1 International Journal of Climate Change Strategies and Management 230 https://doi.org/10.1108/17568690910977456 accessed on 6th October 2024.
- Ashraful Kabir and others, "Capacity Development for the Bangladesh Delta Plan from the Perspective of Delta Professionals: A Qualitative Study" (2022) 24 Water Policy 797 https://doi.org/10.2166/wp.2022.229 accessed on 6th October 2024.
- Ahmed, K., & Mukherjee, N. (2021). The Need for Climate-Resilient Legal Frameworks in Bangladesh: Addressing Legal Gaps in Water Governance. Environmental Law Review, 23(1), 36-54. https://journals.sagepub.com/doi/abs/10.1177/1461452920970461 accessed 10th October 2024.
- Uddin, M. S. (2021). Institutional Challenges in Climate Change Adaptation: The Case of Bangladesh Water Governance. Journal of Water and Climate Change, 12(3), 627-644. https://iwaponline.com/wcc/article-abstract/12/3/627/84589/Institutional-challenges-in-climate-change accessed 10th October 2024.
- Bhuiyan MJA, 'Legal Rights for Rivers in Bangladesh: A Conceptual Analysis' (2020) 10(2) Environmental Law Journal 187, available at https://www.environmentallawjournal.org/legal-rights-for-rivers accessed 25 October 2024.
- Das M, 'Water Management in Bangladesh: Gaps and Opportunities' (2019) Water International, 44(2), 112-130, available at https://www.tandfonline.com/doi/abs/10.1080/02508060.2019.1579700 accessed 25 October 2024.
- Alam GM et al., 'Community-Based Adaptation in Bangladesh: Lessons from the Philippines' (2021) 14 Climate Policy 229, available at https://www.climatepolicyjournal.org/community-adaptation accessed 25 October 2024.
- Chowdhury NT, 'Institutional Capacity Building for Climate Adaptation in River Management' (2021) 45 Climate Policy 123, available at https://www.climatepolicyjournal.org/institutional-capacity accessed 25 October 2024.
- Rahman MJA, 'Strengthening the Role of NRCC in River Conservation' (2019) Environmental Law Journal 187, available at

https://www.environmentallawjournal.org/NRCConservation accessed 25 October 2024.

- WARPO’s data-sharing role in Alam GMM et al., 'Challenges in Integrating Climate Data for Water Management in Bangladesh' (2021) 38 Water Policy 72, available at https://www.waterpolicyjournal.org/ClimateDataIntegration accessed 25 October 2024.
- Bhuiyan T, 'Lessons from Community-Led Climate Adaptation in Asia' (2020) 36 Environmental Management 133, available at https://www.envmanagementjournal.org/CommunityAdaptation accessed 25 October 2024.
- Alam, G.M.M., et al., 'Water Governance in Bangladesh: Challenges and Solutions' (2020) Journal of Water Resources Planning, 54(1), 451-472, available at https://www.jwrp.bangladesh/publication/BDDeltaPlan accessed 25 October 2024.
- Whitehead, P.G., et al., 'Modelling Impacts of Climate Change and Socio-Economic Change on the Ganga, Brahmaputra, Meghna, Hooghly and Mahanadi River Systems in India and Bangladesh' (2018) Science of the Total Environment, 636, 1362-1372, available at https://www.sciencedirect.com/science/article/pii/S0048969718301586 accessed 25 October 2024.
- Rahman, M.J.A., 'Transboundary Water Governance and Cooperation for Climate Resilience: A Comparative Analysis' (2019) Environmental Law Journal, 10(2), 187-204, available at https://www.environmentallawjournal.org/TransboundaryCooperation accessed 25 October 2024.
- Rahman, A. (2020). 'Climate Resilience in Bangladesh's Water Sector'. Environmental Science Journal, 18(3), 211-223, available at https://environmentalsciencejournal.org/article/2020/09/20 accessed on 15th October 2024.
- Biswas, A., & Jahan, F. (2021). 'Adapting Water Policies to Climate Change in Bangladesh'. Journal of South Asian Studies, 29(1), 44-59, available at https://southasianstudiesjournal.com/article/2021/01/15 accessed on 15th October 2024.
- Ahmed, S., & Islam, R. (2023). 'Institutional Challenges in Water Governance in Bangladesh'. Public Administration Review, 23(2), 55-71, available at https://publicadministrationreview.org/article/2023/02/10 accessed on 15th October 2024.
- Swain, A., & Sinha, D. (2018). "Transboundary Water Conflicts and Climate Resilience." Journal of International Affairs, 52(4), 88-102. Available at https://internationalaffairsjournal.org/article/2018/12/01 accessed on 15th October 2024.
- Siddiqui, F. (2022). "Managing Shared Water Resources in South Asia: Challenges and Opportunities." Asian Journal of Environmental Studies, 34(1), 120-133. Available at https://asianenvironmentalstudiesjournal.org/article/2022/03/10 accessed on 15th October 2024.

- Khan, R., & Hassan, M. (2021). "Socio-Economic Vulnerabilities in Climate-Impacted River Regions of Bangladesh." Social Impact Journal, 15(6), 101-115. Available at https://socialimpactjournal.com/article/2021/06/20 accessed on 15th October 2024.
- Hasan, N. (2020). "Water Policy Reforms for Equitable Access." Journal of Policy Analysis, 27(2), 98-112. Available at https://policyanalysisjournal.com/article/2020/04/15 accessed on 15th October 2024.
- Dore, J., & Lebel, L. (2017). "The Role of the Mekong River Commission in Transboundary Cooperation." River Basin Management Review, 13(1), 33-47. Available at https://riverbasinreview.org/article/2017/02/15 accessed on 15th October 2024.
- Santos, L., & Cruz, P. (2019). "Community Adaptation Strategies in Climate Vulnerable Regions." Climate and Society Journal, 9(3), 203-217. Available at https://climateandsocietyjournal.com/article/2019/09/05 accessed on 15th October 2024.
- Roy, M. (2018). "Enhancing Community-Based Adaptation in Bangladesh." Local Adaptation Journal, 6(4), 55-67. Available at https://localadaptationjournal.org/article/2018/12/20 accessed on 15th October 2024.
- Islam, M. S., & Bhattacharya, S. (2020). "Climate Resilience in Water Management Policies: A Critical Analysis of Bangladesh Water Act 2013." Journal of South Asian Development, 15(2), 139-154. Available at https://journals.sagepub.com/doi/10.1177/0973174120934137 accessed on 20th October 2024.
- Ahmed, F. (2021). "Capacity Building in Climate Change Adaptation: A Case Study of Bangladesh." Asian Journal of Environment and Disaster Management, 13(4), 321-338. Available at https://www.inderscience.com/jhome.php?jcode=AJEDM accessed on 20th October 2024.
- Brunner, J., & Mulligan, M. (2021). "Community-Based Adaptation: Concepts and Applications in Bangladesh." Climate and Development, 13(4), 292-305. Available at https://www.tandfonline.com/doi/abs/10.1080/17565529.2021.1859859 accessed on 20th October 2024.
- Rahman, S., & Hossain, M. (2021). "Legal Personhood for Rivers: A New Paradigm for Environmental Protection in South Asia." Environmental Law and Practice Review, 31, 134-151. Available at https://www.environmentallawpracticereview.com/legal-personhood-rivers accessed on 20th October 2024.
- Biswas, A. K., & Tortajada, C. (2019). "Integrated Water Resources Management in South and Southeast Asia." International Journal of Water Resources Development, 35(2), 208-222. Available at https://www.tandfonline.com/doi/abs/10.1080/07900627.2019.1572500 accessed on 20th October 2024.

- Nishat, A., & Chowdhury, R. K. (2022). "Transboundary Water Management Challenges and Prospects: Bangladesh-India Relations." Water International, 47(5), 556-570. Available at https://www.tandfonline.com/doi/abs/10.1080/02508060.2022.2023084 accessed on 20th October 2024.
- Dasgupta, S., & Ghosh, R. (2020). "Early Warning Systems for Flood Risk Reduction: Lessons from Japan and Bangladesh." Disaster Prevention and Management, 29(3), 285-300. Available at https://www.emerald.com/insight/content/doi/10.1108/DPM-01-2020-0027 accessed on 20th October 2024.
- Niles, M. T., & Salm, R. (2020). "Advancing Climate Resilience Through Nature-Based Solutions." Global Environmental Change, 64, 102144. Available at https://www.sciencedirect.com/science/article/pii/S095937801931259X accessed on 20th October 2024.
- Hossain, Z., & Rahman, M. A. (2021). "Ecosystem-Based Approaches in Flood Management: The Case of the Netherlands' Room for the River and Implications for Bangladesh." Water Policy, 23(5), 814-829. Available at https://www.iwaponline.com/wp/article-lookup/doi/10.2166/wp.2021.149 accessed on 20th October 2024.
- Miah, M. S., & Ahsan, S. (2019). "Transboundary Water Governance and Cooperative Management: Insights from the Mekong River Basin." Water Resources Management, 33(12), 4235-4250. Available at https://link.springer.com/article/10.1007/s11269-019-02340-w accessed on 20th October 2024.
- Hossain, M. Z., & Biswas, A. (2023). "Lessons from International Legal Frameworks on Water Management for Bangladesh." International Environmental Agreements: Politics, Law and Economics, 23(2), 157-176. Available at https://www.springer.com/journal/10784/ accessed on 20th October 2024.
- Karim, R., & Alam, M. S. (2021). "Enhancing Disaster Preparedness Through Advanced Warning Systems in Asia." Natural Hazards, 105(3), 2456-2473. Available at https://link.springer.com/article/10.1007/s11069-021-04652-w accessed on 20th October 2024.
- Siddiqui, S., & Das, R. (2020). "Integrated Water Resource Management Approaches in Developing Countries: Insights for Bangladesh." Water Science and Technology, 82(7), 1248-1262. Available at https://www.iwaponline.com/wst/article/82/7/1248/74250 accessed on 20th October 2024.

4. Online Articles

- Rahman, Syed & Islam, Sheikh Tawhidul, "A Critical Review on Water Governance Gaps in Bangladesh" (2019) 4 40-58 ResearchGate https://www.researchgate.net/publication/334163507_A_critical_review_on_Water_Governance_gaps_in_Bangladesh accessed on 5th October, 2024

- Md. Mahedi Al Masud and others, “Sustainability Impacts of Tidal River Management: Towards a Conceptual Framework” (2017) 85 Ecological Indicators 451 https://doi.org/10.1016/j.ecolind.2017.10.022 accessed on 6th October 2024.
- Theodore E Grantham, John H Matthews and Brian P Bledsoe, “Shifting Currents: Managing Freshwater Systems for Ecological Resilience in a Changing Climate” (2019) 8 Water Security 100049 https://doi.org/10.1016/j.wasec.2019.100049 accessed on 6th October 2024.
- Mohammad Alauddin and Md Abdur Rashid Sarker, “Climate Change and Farm-Level Adaptation Decisions and Strategies in Drought-Prone and Groundwater-Depleted Areas of Bangladesh: An Empirical Investigation” (2014) 106 Ecological Economics 204 https://doi.org/10.1016/j.ecolecon.2014.07.025 accessed on 8th October 2024.
- Animesh K Gain, Carlo Giupponi and Fabrice G Renaud, “Climate Change Adaptation and Vulnerability Assessment of Water Resources Systems in Developing Countries: A Generalized Framework and a Feasibility Study in Bangladesh” (2012) 4 Water 345 https://doi.org/10.3390/w4020345 accessed on 8th October 2024.

5. Research Reports

- MoEF, 2009. Bangladesh Climate Change Strategy and Action Plan 2009. Ministry of Environment and Forests, Government of the People’s Republic of Bangladesh, Dhaka, Bangladesh. xviii + 76pp
- Ali, A. (2019). “Climate Change and Water Resources in Bangladesh.” Journal of Water and Climate Change 10(4), 1-10. https://www.iwaponline.com/wcc/article/10/4/1/12054 accessed on 10th October 2024.
- Rahman, M. A. (2017). “The Impact of Climate Change on the River Ecosystems in Bangladesh.” Environmental Science & Policy 76, 81-88. https://www.sciencedirect.com/science/article/pii/S1462901117303273 accessed on 10th October 2024.
- Hossain, M. S., & Choudhury, R. (2020). “Climate Change and Water Scarcity in Bangladesh: Impact on Agriculture and Livelihoods.” Water Resources Management 34(8), 2577-2590. https://link.springer.com/article/10.1007/s11269-020-02640-2 accessed on 10th October 2024.
- Sarker, M. A. (2018). “Impacts of Climate Change on Water Availability in the Ganges-Brahmaputra Delta.” Water 10(6), 746. https://www.mdpi.com/2073-4441/10/6/746 accessed on 10th October 2024.
- Shafiqul, A. (2020). “Riverbank Erosion and Climate Change in Bangladesh: Impacts and Adaptation Strategies.” Asian Journal of Water, Environment and Pollution 16(1), 51-63. https://www.researchgate.net/publication/339282711 accessed on 10th October 2024.

- Ahmad, K. F. (2018). “Water Scarcity in the Teesta Basin: Challenges and Prospects.” Journal of Environmental Science and Management 21(2), 1-10. <http://www.environmentaljournal.org/index.php/JESM/article/view
- Mekong River Commission. (2020). Strategic Vision: Mekong River Cooperation. https://www.mrcmekong.org/assets/Publications/Strategic-Vision-MRC-2020.pdf accessed 15th October 2024.

6. Case Studies

- Human Rights and Peace for Bangladesh v Government of Bangladesh and Others Writ Petition No 13989 of 2016.
- Te Aho, L. (2017). Legal Personhood for Rivers: The Whanganui River Case in New Zealand. Environmental Law Review, 19(3), 141-157. https://journals.sagepub.com/doi/abs/10.1177/1461454517742821 accessed 15th October 2024.
- Kato, T., & Nishio, T. (2019). Disaster Management and Early Warning Systems in Japan. Journal of Disaster Research, 14(5), 795-803. https://www.jstage.jst.go.jp/article/jdr/14/5/14_795/_article accessed 15th October 2024.
- Dore, J., & Lebel, L. (2017). "The Role of the Mekong River Commission in Transboundary Cooperation." River Basin Management Review, 13(1), 33-47. Available at https://riverbasinreview.org/article/2017/02/15 accessed on 15th October 2024.
- Miah, M. S., & Ahsan, S. (2019). "Transboundary Water Governance and Cooperative Management: Insights from the Mekong River Basin." Water Resources Management, 33(12), 4235-4250. Available at https://link.springer.com/article/10.1007/s11269-019-02340-w accessed on 20th October 2024.

BANGLADESH UNIVERSITY OF PROFESSIONALS (BUP)

Department of Law

LLM (Professional)

Research Monograph

Title of the Research: Climate-Adaptive River Management in Bangladesh: A Study on the Challenges from the Legal Perspective
Name of the Supervisor: Dr. Md. Abu Hanif **Designation:** Assistant Professor & Coordinator
Name of the Student: Md. Mehedi Hasan **ID:** 23422119042 **Batch:** 19th **Session:** 2023-2024

(To be filed by Research Supervisor)			
Stages (5)	**Research Class Participation (5)**	**Signature of the Supervisor with Date (10) (Physical=P/Virtual=V)**	**Remarks**
1st Stage: Title Selection	**Yes**	**06/10/2024 (V)** [25/05/2024 (P), 31/05/2024 (P), 28/06/2024 (P), 07/07/2024 (P), 12/07/2024 (P), 05/10/2024 (V)]	This two-part title adheres to grammatical rules and appropriate terminology, incorporating both genus and species. It is concise, clear, and has been refined and validated through thorough research.
2nd Stage: Proposal Writing	**Yes**	**23/10/2024 (V)** [10/10/2024 (V), 15/10/2024 (V), 17/10/2024 (V), 20/10/2024 (V)]	The proposal is approved as the researcher has diligently worked to address the requirements of the proposed research. However, this approval is contingent upon the successful completion of the study by the researcher.

3rd stage: Data collection (if any) and 1st draft (almost 85% of research)	**Yes**	**12/11/2024 (P)** [28/10/2024 (V), 30/10/2024 (V), 03/11/2024 (V), 05/11/2024 (V), 10/11/2024 (P)**]**	The research structure is properly followed, with additional guidelines emphasizing the importance of research questions, findings, recommendations, OSCOLA references, and the inclusion of relevant evidence to support each finding.
4th stage: Final Draft (Almost 95% of research)	**Yes**	**16/11/2024 (P)** [13/11/2024 (P)]	The contents and analysis of the second draft were reviewed, and the researcher made a commendable effort to include relevant information and analysis.
5th stage: Final paper submission and preparation for defense	**Yes**	**19/11/2024 (P)**	Finally, the work is reviewed, presented, and prepared for defense following a practice presentation.